KATHMANDU UNIVERSITY

SCHOOL OF ENGINEERING

DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING

PROJECT PROGRESS REPORT

PERFORMANCE ANALYSIS OF A LORA-BASED COMMUNICATION SYSTEM

A final year project progress report submitted in partial fulfilment
of the requirements for the degree of
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ABSTRACT

This project analyses the performance of a 433 MHz LoRa link between two identical nodes, each consisting of an ESP32 microcontroller, an SX1278 (Ra-01) transceiver and the module's spring helical antenna. The aim is to identify the spreading factor (SF), bandwidth (BW), coding rate (CR) and transmit power that give reliable delivery at the lowest energy, and later to transmit river water-level data using machine-learning-based adaptation.

A simulation framework based only on theory and published values was developed. It models the chirp spread spectrum receiver, time on air and energy, the wireless channel, the Kathmandu University campus in three dimensions with buildings, trees and terrain, and multi-node collisions. The simulated receiver agreed with the SX1278 demodulation limits within 0.4–1.8 dB, and all combinations of SF, BW and CR were found to lie on a single range–energy curve. With antennas 1.5 m high, SF7 served 60 % of the campus at 17 dBm and 22 % at 2 dBm, and choosing the lowest-energy SF and power at each checkpoint saved a median of 76 % of the energy compared with SF9 at 17 dBm.

In the first field session, five campus checkpoints were measured at 2 dBm and compared with the simulation, and a line-of-sight distance test was carried out from 10 m to 180 m. Only the nearest checkpoint, at 51 m, was reliable at every SF, packets were decoded down to the datasheet SNR limits, and the antennas and their installation were found to lose about 18 dB more than assumed. The remaining field sessions and the machine-learning results will be presented in the final report.

Keywords: *LoRa, chirp spread spectrum, wireless channel, spreading factor, three-dimensional propagation model, knife-edge diffraction, packet delivery ratio, field measurement, energy efficiency.*

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SYMBOLS & ABBREVIATIONS

A. Symbols

Table S1: Symbols used in the report

Symbol	Description	Unit	First used in page
SF	Spreading factor	n/a	i
BW	Bandwidth	kHz	i
CR	Coding rate index	n/a	i
M	Number of chips per symbol	n/a	7
T_s	Symbol duration	ms	7
R_b	Equivalent bit rate	bit/s	7
ToA	Time on air of a packet	ms	7
L_p	Path loss	dB	7
d, d_{bp}	Distance, breakpoint distance	m	8
h_t, h_r	Antenna heights	m	8
λ, f	Wavelength, carrier frequency	m, MHz	7
n_1, n_2	Path-loss exponents before and after the breakpoint	n/a	8
X_σ, σ	Shadowing and its standard deviation	dB	8
h(t), K	Fading gain, Rician K-factor	n/a, dB	7
P_t, P_r	Transmitted and received power	dBm	8
G_t, G_r	Antenna gains	dBi	8
N, NF, ΔN	Noise power, noise figure, man-made noise rise	dBm, dB	8
S	Receiver sensitivity	dBm	8
SNR_{req}	SNR required for demodulation	dB	8
P_s	Symbol error probability	n/a	8
G, S_{th}	ALOHA offered load and throughput	n/a	8
E_{pkt}	Energy per packet	mJ	9
I, V	Current, supply voltage	mA, V	9

B. Abbreviations

Table S2: Abbreviations used in the report

Abbreviation	Full form	First used in page
LoRa	Long Range	i
SNR	Signal-to-Noise Ratio	i
ESP32	Espressif 32-bit Wi-Fi/Bluetooth microcontroller	i
CSS	Chirp Spread Spectrum	1
KU	Kathmandu University	1
LPWAN	Low-Power Wide-Area Network	1
PDR	Packet Delivery Ratio	3
RSSI	Received Signal Strength Indicator	3
LoRaWAN	Long Range Wide Area Network (MAC protocol)	5
MAC	Medium Access Control	6
ADR	Adaptive Data Rate	6
ALOHA	Random-access protocol without carrier sensing	6
IoT	Internet of Things	6
LSTM	Long Short-Term Memory	6
CRC	Cyclic Redundancy Check	7
FFT	Fast Fourier Transform	7
SPI	Serial Peripheral Interface	10
PER	Packet Error Rate	13
UTM	Universal Transverse Mercator	13
AWGN	Additive White Gaussian Noise	17
ML	Machine Learning	31

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CHAPTER I: INTRODUCTION

1.1 Background

Remote monitoring systems, such as river water-level, soil-moisture and weather stations, usually send only a few bytes of data every few minutes. They are often installed in places without mains power, Wi-Fi or reliable mobile coverage, such as riverbanks and hilly areas in Nepal. Wi-Fi and Bluetooth reach only a few tens of metres. A cellular modem can cover a long distance, but it uses too much energy for a node that has to run on batteries for several months. Low-Power Wide-Area Network (LPWAN) technologies were developed for this kind of use. They send data slowly, but in return they cover a long distance and use little energy per message [1].

LoRa (Long Range) is one of the most widely used LPWAN technologies. It works in licence-free frequency bands below 1 GHz, such as 433 MHz and 868 MHz. LoRa uses Chirp Spread Spectrum (CSS) modulation, in which each symbol is a chirp whose frequency rises steadily across the channel bandwidth. The receiver matches the whole chirp, so it can decode a packet even when the signal is weaker than the background noise, down to a signal-to-noise ratio (SNR) of about −20 dB [2], [3]. Four settings control a LoRa link: the Spreading Factor (SF), Bandwidth (BW), Coding Rate (CR) and transmit power. Each setting affects range, reliability, time on air and battery use. For example, raising the SF by one step lets the receiver decode a signal about 2.5 dB weaker, but it doubles the time needed to send each packet [4].

Many studies have measured how LoRa performs in real conditions. In open areas with a clear line of sight, links longer than 10 km have been reported, while buildings and trees reduce the range to a much shorter distance [5]. Cattani et al. found that the packet delivery ratio depends on the chosen settings and also on the air temperature. They also found that the settings giving the longest range are not always the ones that use the least energy [6]. Most of these studies used the 868 MHz or 915 MHz band, flat terrain and external antennas. A link at 433 MHz with the small spring antennas of low-cost modules, on a hilly campus with many buildings and trees, may behave quite differently. LoRa has also been evaluated in Nepal [7], but it is still not clear how much each setting affects the link on the Kathmandu University (KU) campus.

In this project, a point-to-point LoRa link at 433 MHz was studied between two identical nodes. Each node consists of an ESP32 microcontroller, an SX1278 transceiver (Ai-Thinker Ra-01) and the spring helical antenna supplied with the module. The work combines a simulation framework, a three-dimensional propagation model of the KU campus and field measurements on the campus. Together, these show how each LoRa setting performs in the place where the system is meant to be used.

1.2 Motivation

Floods are among the most frequent natural disasters in Nepal, and rivers rise rapidly during the monsoon [8]. Many of these rivers flow through remote, hilly areas without mains power or reliable mobile coverage. Water-level monitoring at such sites requires a low-cost link that covers several kilometres and runs on batteries for long periods. A similar need exists on university campuses such as Kathmandu University (KU), where sensor nodes for environmental monitoring, energy metering and water-tank level sensing are spread across buildings, trees and uneven ground that Wi-Fi does not cover reliably.

LoRa offers long range and low energy consumption, which makes it suitable for both cases. However, its performance depends strongly on the link settings and the surrounding environment [5], [9]. A poorly configured node uses a higher spreading factor and transmit power than necessary, which wastes energy and increases collisions with other nodes [10]. A clear understanding of each parameter in real campus and river environments is therefore needed before such systems can be deployed reliably for campus monitoring and early flood warning in Nepal.

1.3 Problem Description

The parameters of a LoRa link involve a trade-off between link strength and cost. A higher spreading factor, a narrower bandwidth, a higher coding rate or a higher transmit power strengthens the link. However, each of these also increases the time on air and the energy per packet, and reduces the capacity of the network. The most suitable setting therefore differs from one node to another. In low-cost deployments, the antennas are usually mounted only 1.5–2 m above the ground. At this height, the received power is strongly affected by ground reflection, walls, trees and terrain, so general path-loss models cannot reliably predict the spreading factor required at a given location in a built-up, hilly area. The actual gain of the small spring antennas supplied with low-cost modules and the

background noise level at 433 MHz are also not well documented, and must be measured in the field. In addition, the channel varies with the movement of people and vehicles, with rain and with temperature. A fixed setting is therefore either unreliable when conditions worsen or wasteful when conditions improve, and an adaptive method of selecting the link parameters is required.

1.4 Objectives

The objectives of this project are:

- To evaluate the effect of SF, BW, CR and transmit power on RSSI, SNR, PDR and energy, through simulation and field measurement at KU.
- To model the KU propagation environment and calibrate the model with field data.
- To transmit ultrasonic water-level data reliably using machine-learning-based adaptive SF selection at the lowest workable transmit power.

1.5 Methodology

The work was carried out in four stages. First, a simulation framework was developed in Python using theory and published values only. It models the LoRa physical layer, the wireless channel and a multi-node network, together with a three-dimensional model of the KU campus with buildings, tree canopy and terrain elevation. Second, the ESP32 firmware was modified so that the transmitter steps through SF7 to SF12 automatically and the receiver follows the same sequence. Field measurements were then taken on the campus with both antennas mounted on 1.5 m tripods. The RSSI, SNR, noise floor and PDR were logged on a laptop at the checkpoints and at a 10 m line-of-sight reference point. Third, the measured received power was compared with the 3D model, and a first estimate of the antenna loss was fitted. The path-loss slope, obstruction loss and noise level are being refined as further checkpoints are measured. Finally, a machine-learning pipeline for selecting the SF and transmit power was developed and tested on simulated data. It is to be retrained on the measured data and applied to the river water-level node.

1.6 Limitations

The limitations of this project are:

- Only one transmitter and one receiver at 433.175 MHz were used. Multi-node performance was studied by simulation only.
- The 3D propagation model covers only the mapped KU campus; ranges beyond it (up to 2 km) were estimated with a general distance model.
- Wall loss, foliage loss and building heights were assumed in the model. The five measured checkpoints show the overall difference between the model and the measurements, but they are not enough to determine each of these losses separately.
- Interference from individual sources on the campus, such as other devices that may use the 433 MHz band and noise from nearby electronic equipment, was not identified or modelled. Only an average man-made noise level was assumed in the simulation.

1.7 Organisation of Report

The report is divided into five chapters. Chapter I gives the background, motivation, problem, objectives, methodology and limitations of the project. Chapter II reviews previous work on LoRa performance, propagation, network capacity, adaptive data rate and machine learning, and presents the equations used in the simulation. Chapter III describes the nodes, the circuit, the simulation framework, the field-measurement procedure and the machine-learning method. Chapter IV discusses the simulation results and the first field session, and Chapter V gives the conclusions and future work. The appendices contain the pin connections, hardware photographs, checkpoint data and main program code.

1.8 Summary

This chapter introduced LoRa as a low-power, long-range technology for remote monitoring and explained the need to study its parameters on the KU campus. The range and energy of a LoRa link depend on the SF, BW, CR, transmit power, antennas and surroundings, which are difficult to predict on a hilly campus with buildings and trees. The project therefore combines a simulation framework, a three-dimensional campus model and field measurements, aiming at adaptive SF and power selection for a river water-level node. The objectives, the methodology and the main limitations of the work are also presented.

CHAPTER II: TECHNOLOGY AND LITERATURE SURVEY

This chapter reviews the previous work related to this project and presents the theory and equations used in the simulation.

2.1 Literature Review

LPWANs were described by Centenaro *et al.* [1] and Raza *et al.* as star networks that give up data rate in exchange for long range and long battery life. Among the LPWAN technologies, LoRa is widely used for research and small private networks because it works in the unlicensed sub-GHz bands and the radio modules are inexpensive. The LoRa physical layer and the LoRaWAN medium access protocol were explained by Augustin *et al.* [4] and Haxhibeqiri *et al.* who showed that SF, BW and CR together set the data rate and the receiver sensitivity. The reference equations for time on air and the SNR limits of each SF are given in the Semtech application notes [2] and the SX1276/77/78/79 datasheet [3]. Vangelista [11] described LoRa as frequency-shift chirp modulation demodulated with a discrete Fourier transform, and Elshabrawy and Robert derived an approximate bit error rate for this receiver. These works were used to build and validate the modulation model of this project.

Several field studies have shown how much the LoRa range depends on the site. Petäjäjärvi *et al.* [5] measured coverage at 868 MHz in Oulu, Finland, with the base-station antenna on a 24 m tower, and reported links of more than 15 km on ground and close to 30 km over water, together with a path-loss model for the area. Bor *et al.* tested different LoRa settings and showed that the link quality depends on the combination of SF, BW, CR and power. In a long-term experiment, Cattani *et al.* [6] found that a higher SF does not always improve delivery and that high temperature lowers the received power of SX127x radios. Iova *et al.* [9] compared city and mountain deployments and found that vegetation reduces the range considerably. González-Palacio *et al.* published a LoRaWAN path-loss dataset in which environmental variables improved the prediction. In Nepal, Pokharel *et al.* [7] evaluated LoRa nodes for rural connectivity in terms of RSSI, SNR and packet loss. Kaur *et al.* [12] deployed a 433 MHz LoRa network on a university campus and found that the best SF and CR differed between line-of-sight and obstructed areas; on line-of-sight paths, SF9 with CR 4/5 gave an RSSI of −89.12 dBm and an SNR of 9.91 dB. In most of these studies, the gateway was mounted high or a high-gain antenna was used. Links between two nodes less

than 2 m above the ground with small module antennas, as in this project, are rarely reported.

For propagation modelling, the log-distance model with log-normal shadowing, the two-ray ground-reflection model and the Rayleigh and Rician fading models are commonly used in link-budget studies [13]. Free-space loss is given in ITU-R P.525. The loss through buildings can be estimated with the COST 231 multi-wall model and the material losses of ITU-R P.2040 [14], while the loss through trees is usually estimated with the Weissberger model, which is valid from 230 MHz to 95 GHz [15]. ITU-R P.372 gives the man-made noise level above thermal noise in residential and business areas [16]. These models were combined in the three-dimensional campus model of this project.

The capacity of LoRa networks has been studied mainly by analysis and simulation. Adelantado *et al.* [17] showed that pure ALOHA access and the duty-cycle limits restrict the capacity of LoRaWAN. Georgiou and Raza found that the coverage probability falls quickly as the number of nodes increases, especially at high SF where packets are long. Bor *et al.* [10] developed the LoRaSim simulator and showed the benefit of the capture effect and of selecting the SF for each node. Croce *et al.* [18] measured the signal-to-interference thresholds between different SFs and showed that they are not perfectly orthogonal, and Mahmood *et al.* included this effect in a scalability analysis. The adaptive data rate (ADR) of LoRaWAN reduces the SF and power of a node when the SNR of its recent packets is high [19]. However, Slabicki *et al.* and Kufakunesu *et al.* reported that the standard ADR reacts slowly when the channel changes. Simulators such as the ns-3 LoRaWAN module and LoRaWANSim model the physical layer, the MAC layer and energy separately, and Casals *et al.* [20] and Bouguera *et al.* estimated the energy consumption of LoRa nodes from the current in each operating state.

Machine learning has recently been applied to the selection of LoRa parameters. Nisar *et al.* [21] used a long short-term memory (LSTM) network to allocate spreading factors and reported a higher PDR and about 30 % lower energy consumption than the baseline methods. Most such models were trained and tested on simulated data or on data from a single site. For the target application, Kabi *et al.* [22] developed a low-cost open-hardware LoRa node for river water-level monitoring, which shows that the application is practical. In Nepal, the Radio Frequency Policy 2080 allocates 865–868 MHz to machine-to-machine and IoT devices [23], while for 433.05–434.79 MHz the European ERC Recommendation 70-03 allows only low power and a limited duty cycle [24].

From the review, it was found that LoRa parameters and propagation have been studied widely, but mostly at 868 MHz or 915 MHz, with elevated gateways and on flat sites. Few studies have considered low 433 MHz links with small module antennas on a hilly campus with buildings and trees, a site-specific model calibrated with measurements from the same site, or the use of such data to select the SF and power adaptively. This project addresses these points.

2.2 Theoretical Background

2.2.1 LoRa Modulation and Parameters

A LoRa symbol carries SF bits and is one of $M = 2^{SF}$ cyclic shifts of a base up-chirp that sweeps across the bandwidth BW [2], [11]. Symbol *s*, sampled at the chip rate, is

$$x_s[n] = \exp(j\pi (k^2/M - k)), \quad k = (n + s) \bmod M, \quad n = 0, 1, \ldots, M - 1 \tag{1}$$

The receiver multiplies the samples by the conjugate of the base chirp, which turns the symbol into a single tone, and finds the tone with an FFT:

$$\hat{s} = \arg\max_m \left| \Sigma_n \, r[n] \cdot x_0^*[n] \cdot \exp(-j2\pi mn/M) \right| \tag{2}$$

The FFT gives a processing gain of $10 \log_{10}(M)$ dB. Each increase in SF therefore lowers the required SNR by about 2.5 dB, from −7.5 dB at SF7 to −20 dB at SF12 [3]. The symbol time, bit rate and time on air are [3]

$$T_s = 2^{SF} / BW, \quad R_b = SF \cdot (4 / (4 + CR)) \cdot BW / 2^{SF} \tag{3}$$

$$n_{pay} = 8 + \max(\lceil A / (4(SF - 2DE)) \rceil \cdot (CR + 4), 0) \tag{4}$$

$$ToA = (n_{pre} + 4.25) \cdot T_s + n_{pay} \cdot T_s \tag{5}$$

where $A = 8PL - 4SF + 28 + 16CRC - 20IH$, PL is the payload in bytes, CR = 1 to 4 for coding rates 4/5 to 4/8, $n_{pre} = 8$, CRC = 1, IH = 0 for an explicit header and DE = 1 when T_s exceeds 16 ms. One step in SF doubles T_s and approximately doubles the time on air; halving the bandwidth has the same effect.

2.2.2 Wireless Channel and Link Budget

The received baseband signal of a narrowband link can be written as [13]

$$r(t) = \sqrt{(10^{-Lp(d)/10})} \cdot h(t) \cdot x(t) + n(t) \tag{6}$$

where $L_p(d)$ is the path loss including shadowing, h(t) is the fading gain with $E[|h|^2] = 1$ and n(t) is the noise. The free-space loss is $L_{FS} = 20 \log_{10}(4\pi d/\lambda)$. For antennas close to the

ground, the ground-reflected ray makes the loss increase faster beyond the breakpoint distance $d_{bp} = 4h_t h_r/\lambda$, which is about 13 m for 1.5 m antennas at 433 MHz [13]. A dual-slope model with $n_1 = 2$ up to d_{bp} and $n_2 = 3$ beyond it was used:

$$L_p(d) = L_{FS}(1\ \text{m}) + 10\, n_1 \log_{10}(\min(d, d_{bp})) + 10\, n_2 \log_{10}(\max(d/d_{bp}, 1)) + X_\sigma \qquad (7)$$

Walls add a fixed loss each [14], trees add the Weissberger loss [15], and roofs or ridges in the path add knife-edge diffraction loss according to ITU-R P.526 [25]. The shadowing X_σ is Gaussian in dB with a standard deviation of about 3 dB on line-of-sight paths and 6–8 dB on obstructed paths. Fading is Rician (K = 6–9 dB) when a direct path exists and Rayleigh otherwise. Since campus delay spreads of 0.1–1 µs give a coherence bandwidth much wider than 62.5 kHz, the fading is flat. A walking person (1.4 m/s) gives a coherence time of about 0.2 s, shorter than an SF12 packet, so all measurements were taken with the receiver stationary.

The noise power in the channel and the receiver sensitivity are

$$N = -174 + 10 \log_{10}(BW) + NF + \Delta N \ \ [\text{dBm}], \quad S = N + SNR_{req}(SF) \ \ [\text{dBm}] \qquad (8)$$

where NF = 6 dB is the noise figure and ΔN is the man-made noise above thermal noise [16]. The received power and the link margin are

$$P_r = P_t + G_t + G_r - L_p(d), \quad M_{link} = P_r - S \qquad (9)$$

The Ra-01 spring antenna is much shorter than $\lambda/4 = 173$ mm, so its gain is expected to be negative [26]. A gain of −5 dBi per antenna and a mismatch loss of 1 dB were assumed.

2.2.3 Packet Error, Collisions and Energy

LoRa uses a Hamming code with a diagonal interleaver, so one symbol error causes at most one bit error per codeword. For a symbol error rate P_s and N_{blk} blocks of (4 + CR) symbols, the PDR at CR 4/7 is

$$PDR = [\ (1 - P_s)^{4+CR} + (4 + CR)\, P_s\, (1 - P_s)^{3+CR}\ ]^{Nblk} \qquad (10)$$

Nodes share the channel without carrier sensing (pure ALOHA), so the throughput for an offered load G is [17]

$$S_{th} = G \cdot e^{-2G} \qquad (11)$$

A packet survives a collision on the same SF if it is at least 6 dB stronger (capture effect) and a collision on another SF if it exceeds the thresholds of −8 dB to −25 dB measured in [18]. The energy per packet is

$$E_{pkt} = V \cdot (I_{TX}(P_t) + I_{MCU}) \cdot ToA \quad (12)$$

where I_{TX} is the SX1278 current (87 mA at 17 dBm [3]), $I_{MCU} \approx 30$ mA is the ESP32 current, V is the supply voltage and ToA is the time on air [20].

2.3 Summary

The literature review showed that LoRa trades data rate for range through the SF, BW and CR, and that the achieved range depends strongly on antenna height, terrain and vegetation. Most published measurements were made at 868 MHz or 915 MHz with elevated gateways on flat sites, while low 433 MHz links with small antennas on a hilly campus have received little attention. Network studies showed that long packets at high SF limit capacity and that the standard ADR adapts slowly, which has led to machine-learning approaches for SF selection. The theory used in the simulation was then presented, including the chirp modulation and time on air, the dual-slope path loss with wall, foliage and diffraction losses, the noise and link budget, and the packet error, collision and energy models.

CHAPTER III: METHODOLOGY

This chapter describes the hardware of the transmitting and receiving nodes, the simulation framework, the field-measurement procedure and the machine-learning method.

3.1 Transmitting Node

The transmitting node consisted of an ESP32 microcontroller, an SX1278 LoRa module (Ai-Thinker Ra-01) with its spring helical antenna, and a power supply, as shown in Figure 3.1. The ESP32 controlled the LoRa module through the SPI interface using the RadioLib library [27], [3]. For the field test, the node was programmed as a beacon that steps through SF7 to SF12 and sends 20 numbered packets at each setting.

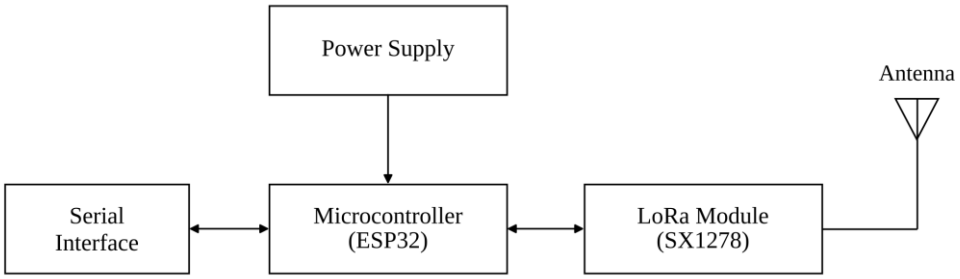


Figure 3.1: Block diagram of the transmitting node

3.2 Receiving Node

The receiving node used the same ESP32 and SX1278 hardware with a power supply and a laptop, as shown in Figure 3.2. The ESP32 followed the parameter sequence of the transmitter, decoded each packet and sent its sequence number, RSSI, SNR, frequency error and CRC status to the laptop over USB. The noise floor and the PDR of each 20-packet window were also reported.

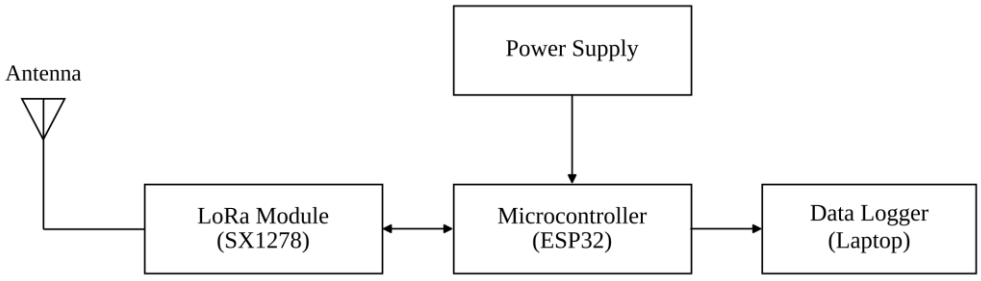


Figure 3.2: Block diagram of the receiving node

3.3 Schematic Diagram

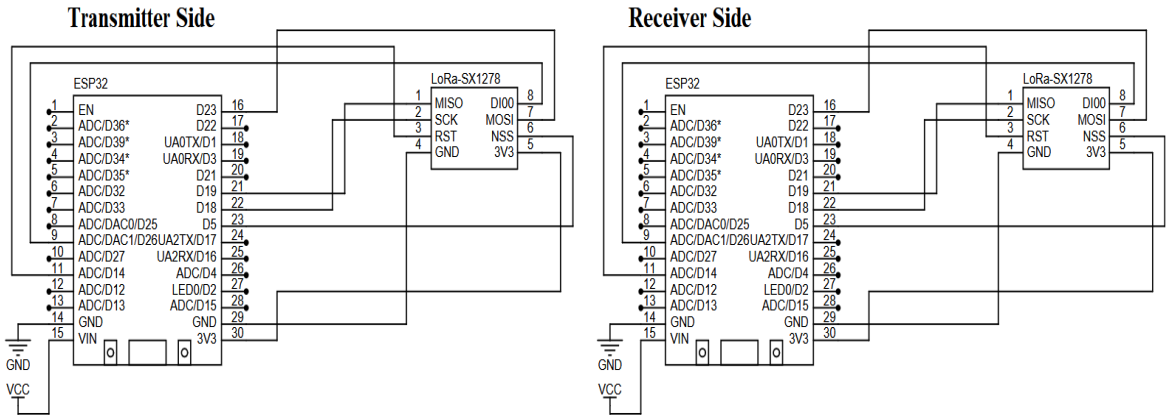


Figure 3.3: Schematic diagram

Figure 3.3 shows the circuit of the transmitter and the receiver. The SX1278 module is connected to the ESP32 through the SPI lines (SCK, MISO, MOSI and NSS), the RESET line and the DIO0 interrupt line, which signals the end of a transmission or the arrival of a packet. The module is powered from the 3.3 V regulator of the ESP32 board and shares a common ground with it. The pin assignments are listed in Appendix A.1, the components in Appendix A.2 and photographs of the setup in Appendix A.3.

3.4 Simulation Framework

The simulation was written in Python using NumPy, SciPy, pandas and Matplotlib. Only theory and published values were used, so that the simulation gives an independent prediction that can be compared with the field data. The steps are shown in Figure 3.4, and the parameters and assumptions are listed in Table 3.1. The antenna height was set to 1.5 m at both ends, equal to the height of the tripods used in the field test.

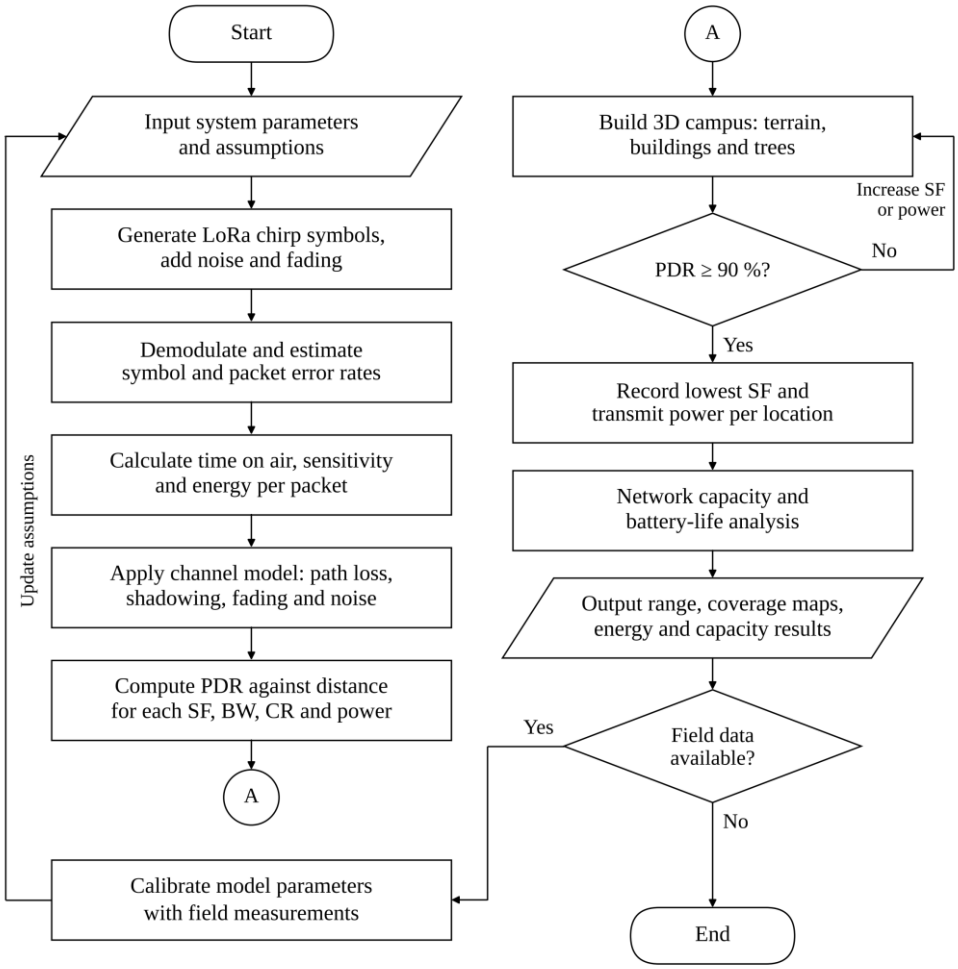


Figure 3.4: Flowchart of the simulation

Table 3.1: Simulation parameters

Quantity	Value used	Basis
Carrier frequency	433.175 MHz (λ = 0.692 m)	Design
Transmit power	2–17 dBm (17 dBm operating, 2 dBm field test)	[3]
Main configuration	BW 62.5 kHz, CR 4/7, 20-byte payload, 8-symbol preamble, CRC on	Design
Demodulation SNR limit	−7.5 dB (SF7) to −20 dB (SF12)	[3]
Noise figure and man-made noise	6 dB; 8 dB rise on campus, 1–2 dB in vegetation	[3], [16]
Antenna gain	−5 dBi per antenna and 1 dB mismatch	Assumption [26]
Antenna height	1.5 m at both ends (tripods)	Design
Line-of-sight path loss	n = 2 up to the breakpoint (13 m at 1.5 m), n = 3 beyond	[13]
Building heights	1, 2 or 3 storeys of 3.5 m from the footprint area	Assumption
Tree canopy	8 m high; 40 % of the mapped green area is tree foliage	Assumption, [15]
Ground elevation	Copernicus GLO-30 elevation model (30 m grid)	[28]

Quantity	Value used	Basis
Obstruction losses	7 dB per wall; multiple knife-edge diffraction; 30 dB for the path around buildings	[14], [25], assumption
Supply current	SX1278: 24 mA at 2 dBm to 87 mA at 17 dBm; ESP32: 30 mA	[3]
Collisions	6 dB co-SF capture and measured inter-SF thresholds	[10], [18]

In the Monte-Carlo simulation of the modulation, random chirp symbols were generated with Equation (1), white Gaussian noise was added and the symbols were decoded with Equation (2). For each SF, the SNR was increased in 1 dB steps around the datasheet limit until about 200 symbol errors were counted at each point. The simulation was repeated with Rayleigh and Rician fading, and the symbol error rates were used in the packet error model of Section 2.2.3. The PDR at distance *d* was then averaged over 1500 realisations of shadowing and fading:

$$\mathrm{PDR}(d) = \mathrm{E}\left[\, 1 - \mathrm{PER}\left(P_t + G_t + G_r - L_p(d) - X_\sigma + 10 \log_{10} g - N \right) \right] \tag{13}$$

The range of a configuration was taken as the largest distance with a PDR of at least 90 %. Distances up to 2 km were evaluated in three generic campus environments for all combinations of SF, BW, CR and transmit power. Each uncertain assumption was also varied between a pessimistic and an optimistic value to find its effect on the results.

For the campus model, the KU topographic map (1:1250, UTM zone 45N) was digitised and georeferenced with its 100 m grid, giving 0.87 m per pixel, and a drone photograph was registered to it with an error of about 4 m. Each pixel was classified as building, vegetation, open ground, road or another class from the legend colours. The model was then extended to three dimensions, as shown in Figure 3.5. Building footprints were extruded to one, two or three storeys according to their area, vegetation was given an 8 m canopy, and the ground elevation was taken from the Copernicus 30 m elevation model [28]. Since this is a surface model that includes roofs and tree tops, these were removed with a morphological opening filter about 90 m wide to obtain the bare ground. The transmitter was placed on the plaza south of the upper fountain roundabout (27.61857 N, 85.53858 E) with its antenna 1.5 m above the ground.

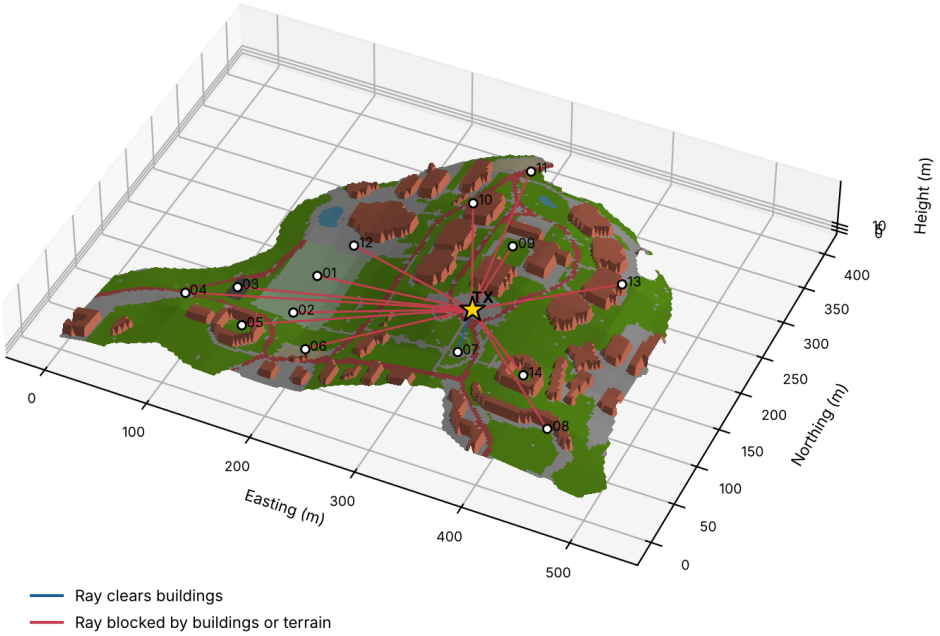


Figure 3.5: 3D model of the KU campus

For each point on the campus, the straight ray between the antennas was compared with the height profile of the ground, roofs and trees, and the path loss was calculated as

$$L = L_{dual}(d_{3D}) + L_{ant} + L_{obs} + L_{veg} \quad [dB] \tag{14}$$

$$L_{obs} = -10 \log_{10}(10^{-L_{pen}/10} + 10^{-L_{dif}/10} + 10^{-L_{sc}/10}) \tag{15}$$

where L_{dual} is the dual-slope loss of Equation (7) and L_{ant} = 11 dB is the antenna loss. The obstruction loss L_{obs} adds in power three paths past a building: through the walls (L_{pen}, 7 dB per wall and 0.3 dB/m inside), over the roofs and ridges by diffraction (L_{dif}), and around the building by scattering (L_{sc}, assumed 30 dB). L_{dif} was calculated with the Deygout method for up to three knife edges each given by [25]

$$J(v) = 6.9 + 20 \log_{10}(\sqrt{((v - 0.1)^2 + 1)} + v - 0.1), \quad v = h \sqrt{(2(d_1 + d_2) / (\lambda\, d_1\, d_2))} \tag{16}$$

where h is the height of the edge above the ray and d_1, d_2 are its distances from the two ends. L_{veg} is the Weissberger loss for the part of the ray below the tree canopy. The lowest SF giving a PDR of 90 % was found for every point at transmit powers from 2 dBm to 17 dBm. Fourteen checkpoints were selected on open ground, between buildings, among trees and at the edges of the campus; their coordinates are given in Appendix B.1.

The network was simulated as pure ALOHA. A packet was lost if it failed the error check of Equation (10) or collided with a packet that was not weaker by the capture threshold.

Fixed SF7, SF9 and SF12 were compared with a per-node rule that gives each node the lowest SF with at least 5 dB SNR margin, similar to LoRaWAN ADR [19].

3.5 Field Measurement Procedure

The field test was designed to check the simulation directly, following Figure 3.6. The transmitter was placed at the campus centre on a 1.5 m tripod and set to 2 dBm, because at 17 dBm the simulation predicts SF7 for most of the campus and the difference between the SFs would not be seen. Two complete SF sweeps were recorded at each checkpoint with the receiver on a 1.5 m tripod. A separate line-of-sight distance test from 10 m to 180 m was carried out with SF9, the library default output of 17 dBm and the antennas about 2 m above the ground. Every packet and every 20-packet window were stored on the laptop (Appendix B.2), and the PDR was calculated as

$$\text{PDR}\ (\%) = (N_{\text{received}} / N_{\text{sent}}) \times 100 \qquad (17)$$

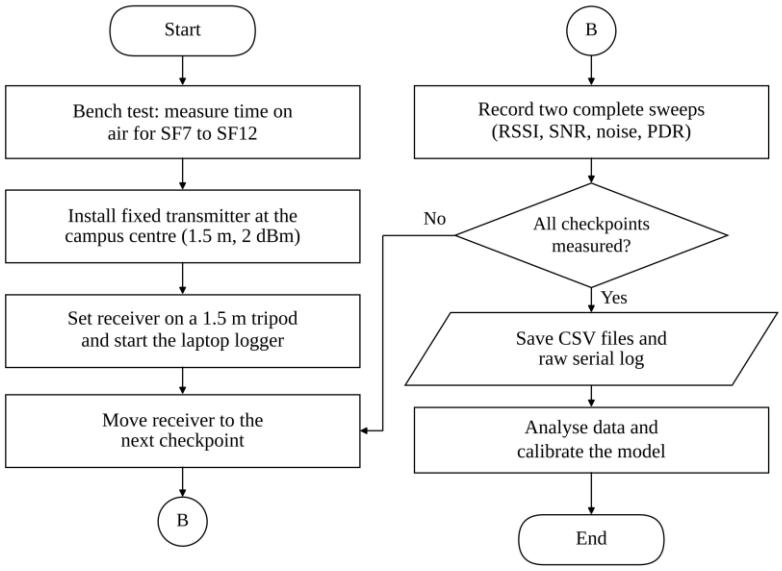


Figure 3.6: Flowchart of the field measurement

The measured received power at each checkpoint was compared with the campus model under the same conditions, and the antenna loss was fitted by least squares.

3.6 Machine-Learning Method

For the adaptive part, each 10-packet window received at the current SF forms one sample. The features are the fraction of packets received, the mean, minimum and spread of SNR and RSSI, the noise floor and the SNR margin above the limit of the current SF. The label is the lowest SF that reached a PDR of 90 % in the other sweep of the same visit. Location

and distance are not used, so that the model can be applied to other environments. Decision tree, random forest, support vector machine, k-nearest-neighbour and gradient-boosting classifiers will be compared with the LoRaWAN ADR rule [19], and the selected model will be implemented on the ESP32. The main parts of the firmware and the simulation programs are listed in Appendix C.

3.7 Summary

This chapter described the transmitting and receiving nodes, which use an ESP32 and an SX1278 module with spring antennas, and the circuit connecting them. The simulation framework was then explained. It includes a Monte-Carlo model of the LoRa modulation, a link model with path loss, shadowing and fading, a three-dimensional model of the KU campus with buildings, trees and terrain, and a network model. The field test was planned at 2 dBm with the transmitter at the campus centre and 14 checkpoints, a line-of-sight distance test was carried out, and the received power was compared with the campus model. Finally, the machine-learning method for selecting the SF from measured link quality was described.

CHAPTER IV: RESULTS AND ANALYSIS

This chapter presents the simulation results in Section 4.1 and the results of the first field session, compared with the simulation, in Section 4.2. The chapter ends with a summary and the status of the machine-learning work.

4.1 Simulation Results

The simulation results are presented in the order of the flowchart in Figure 3.4. They were obtained from theory and published values only and are compared with the field measurements in Section 4.2.

4.1.1 Validation of the LoRa Physical-Layer Model

Figure 4.1 shows the symbol error rate of the simulated chirp receiver in white Gaussian noise. The curves are separated by about 2.5 dB per SF, as expected from the processing gain $10 \log_{10}(2^{SF})$, and each curve falls steeply close to the SX1278 datasheet limit shown by the dotted lines. Table 4.1 compares the SNR at which the complete packet model (20 bytes, CR 4/7 and 1 dB implementation loss) reaches 1 % packet error with the datasheet limit. The two agree within 0.4 dB at SF7 and 1.8 dB at SF12. In the presence of fading the error rate falls much more slowly, so shadowing and fading were included in all link and campus results.

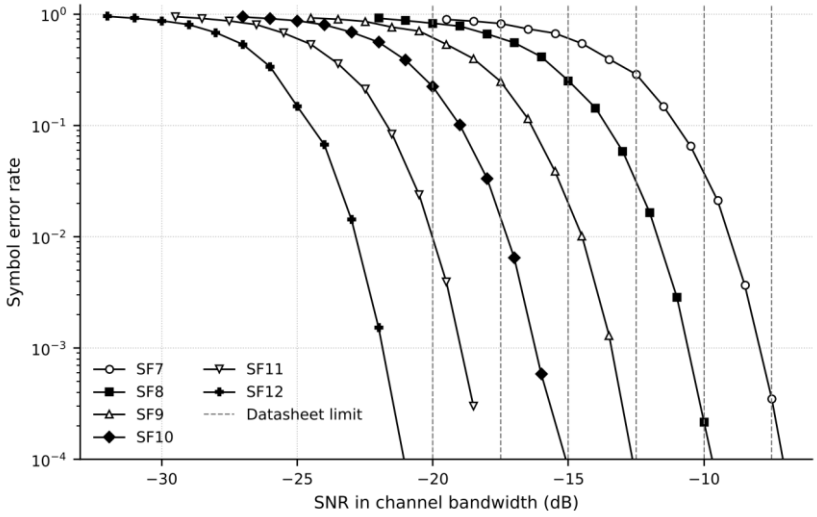


Figure 4.1: Symbol error rate in AWGN

Table 4.1: SNR for 1 % packet error: simulation and datasheet

SF	SF7	SF8	SF9	SF10	SF11	SF12
Datasheet limit (dB)	-7.5	-10.0	-12.5	-15.0	-17.5	-20.0
Model, PER 1 % (dB)	-7.9	-10.6	-13.4	-16.1	-18.9	-21.8
Difference (dB)	-0.4	-0.6	-0.9	-1.1	-1.4	-1.8

4.1.2 Time on Air, Range and Energy

Table 4.2 lists the physical-layer quantities of the main configuration. Each increase in SF doubles the symbol time and roughly doubles the time on air and the energy, while it improves the sensitivity by 2.5 dB. Reducing the transmit power from 17 dBm to 2 dBm lowers the energy per packet by a factor of 2.2 at every SF. A 20-byte packet at SF12 occupies the channel for 3.16 s, so a 1 % duty-cycle limit allows only 11 packets per hour, which is fewer than the 12 needed for one report every 5 minutes [24].

Table 4.2: Time on air, sensitivity and energy for each SF

SF	T_s (ms)	Time on air (ms)	Bit rate (bit/s)	Sensitivity (dBm)	Energy (mJ)
SF7	2.05	141.8	1953	-127.5	54.8
SF8	4.10	255.0	1116	-130.0	98.4
SF9	8.19	452.6	628	-132.5	174.8
SF10	16.38	1019.9	349	-135.0	393.8
SF11	32.77	1810.4	192	-137.5	699.0
SF12	65.54	3162.1	105	-140.0	1220.9

Figure 4.2 shows the PDR against distance between campus buildings at 17 dBm, obtained with path loss, shadowing ($\sigma = 7$ dB) and Rayleigh fading. Table 4.3 summarises the range at which the PDR remains at or above 90 %. Between buildings, the range increases from 475 m at SF7 to 1.4 km at SF12, while through trees SF7 already reaches 1.5 km and on a clear line of sight every SF exceeds the 2 km limit of the study.

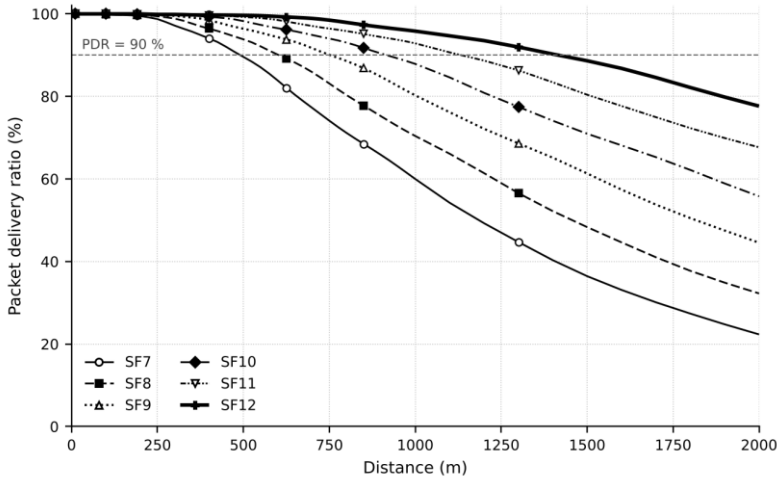


Figure 4.2: PDR against distance between buildings

Table 4.3: Range at 90 % PDR for each SF

Environment	SF7	SF8	SF9	SF10	SF11	SF12
Campus line of sight	> 2000	> 2000	> 2000	> 2000	> 2000	> 2000
Between buildings	475	600	750	900	1100	1400
Through trees	1500	1900	> 2000	> 2000	> 2000	> 2000

Figure 4.3 plots range against energy per packet for 72 combinations of SF, BW and CR. The points lie close to a single curve, so additional range costs about the same energy whichever parameter is changed. Table 4.4 shows the effect of changing one parameter at a time from the baseline of SF9, 62.5 kHz, CR 4/7 and 17 dBm. The SF has the strongest effect on both range and energy, while the coding rate changes the range by less than 10 %, so CR 4/7 was kept fixed.

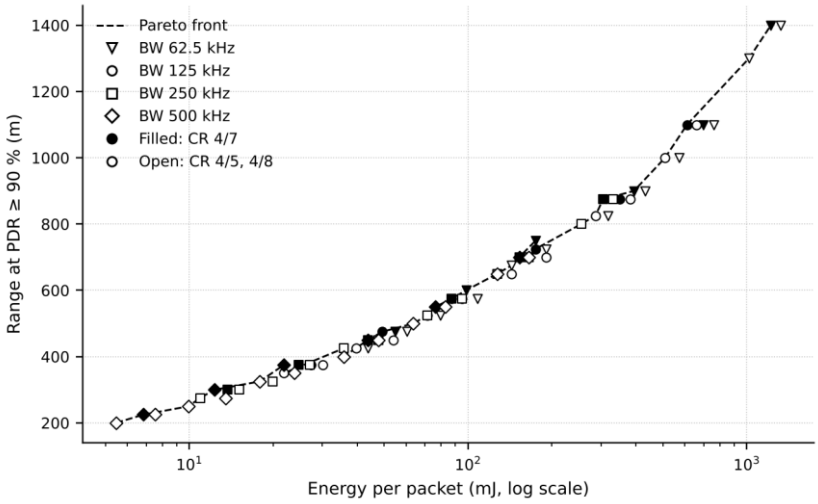


Figure 4.3: Range against energy for all SF, BW and CR combinations

Table 4.4: Effect of changing one parameter

Parameter	Setting	Range (m)	Range change	Energy (mJ)	Energy change
SF	SF7	475	-37 %	54.8	-69 %
SF	SF12	1400	+87 %	1220.9	+599 %
Bandwidth	125 kHz	575	-23 %	87.4	-50 %
Bandwidth	250 kHz	450	-40 %	43.7	-75 %
Coding rate	CR 4/5	675	-10 %	143.1	-18 %
Coding rate	CR 4/8	725	-3 %	190.6	+9 %
TX power	2 dBm	225	-70 %	80.7	-54 %
TX power	11 dBm	475	-37 %	101.6	-42 %

4.1.3 KU Campus Coverage

Of the 13.6 ha of mapped campus, 47 % is covered by vegetation and 16 % by buildings, so most paths from the transmitter meet trees, walls or both. Figure 4.4 shows one such path, from the transmitter to CP10, in side view. With both antennas at 1.5 m, the ray passes below the roofs of the buildings on the path and through several tree canopies. The elevation model also shows the ground rising by about 5 m between 20 m and 60 m from the transmitter, towards the higher central courtyard, so the direct ray and most of the first Fresnel zone are blocked by the ground itself. The signal therefore reaches CP10 through the walls, over the roofs by diffraction, or around the buildings, and the model combines these three paths in power.

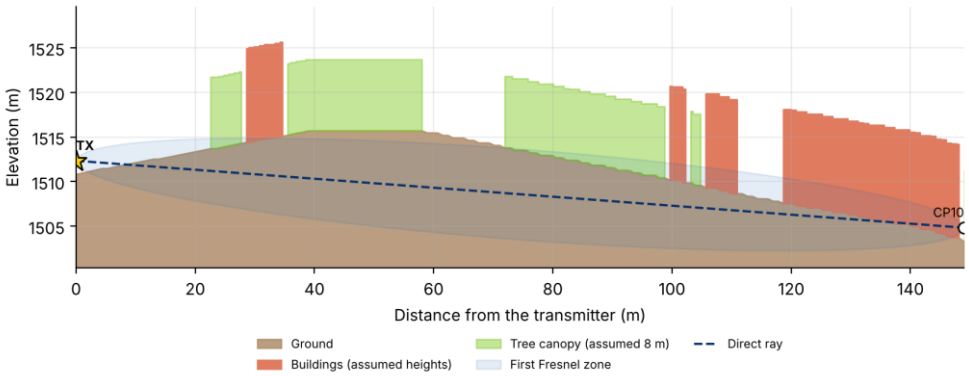


Figure 4.4: Path profile from the transmitter to CP10

Figure 4.5 shows the lowest SF needed at each point at the field-test power of 2 dBm. Close to the transmitter SF7 is sufficient, but radial shadows extend behind each group of buildings, and 44 % of the campus has no link at 2 dBm. Most of this area lies in the lower western part of the campus, around the football ground and the basketball court, about 30

m below the transmitter, where the edge of the higher ground blocks the path, and in the outer areas behind several blocks. Figure 4.6 extends the result to all transmit powers. The share of the campus served at SF7 rises from 22 % at 2 dBm to 60 % at 17 dBm, the area without a link fall from 44 % to 4 %, and the mean time on air of the required SF falls from 1067 ms to 349 ms.

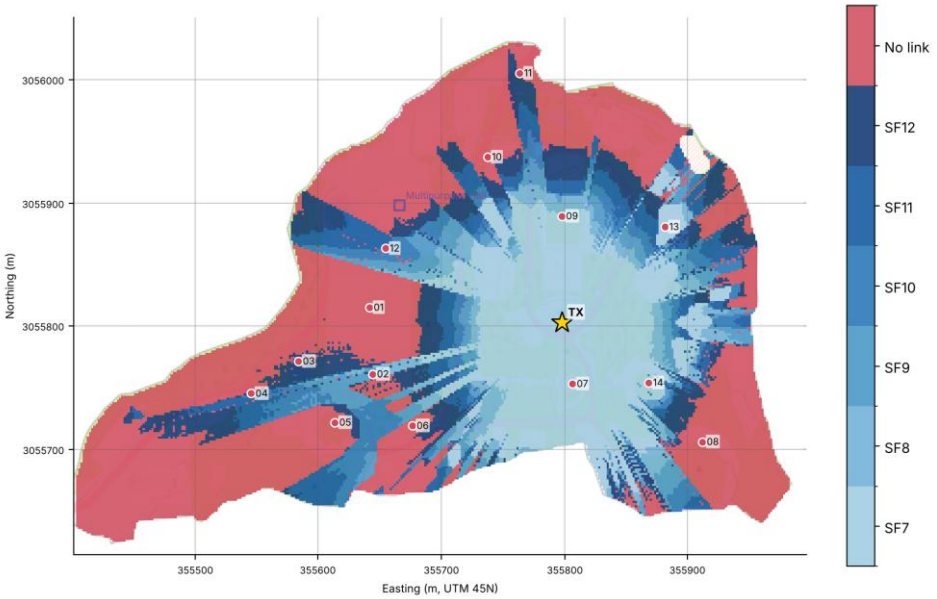


Figure 4.5: Lowest SF needed over the campus at 2 dBm

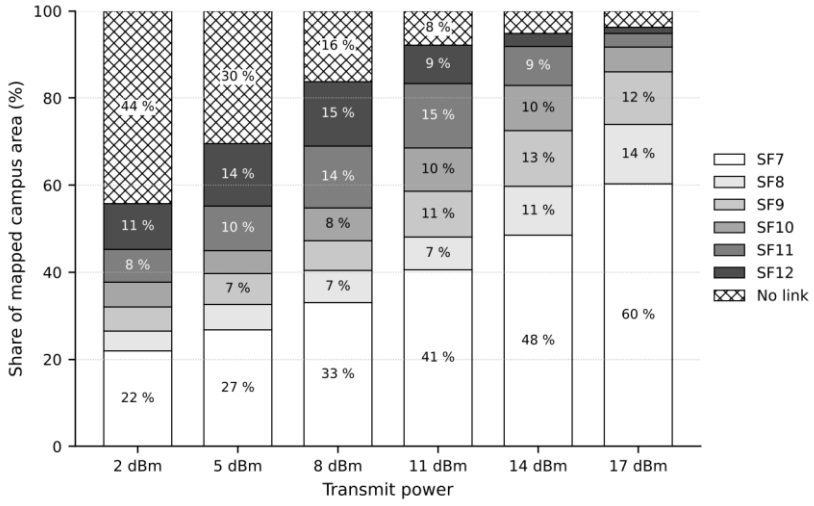


Figure 4.6: Share of the campus needing each SF

Figure 4.7 shows how the share of the campus served at SF7 at 2 dBm changes when each uncertain assumption is set to a pessimistic and an optimistic value. The spring antenna gain (each) has the largest effect (16 % to 33 %), followed by the man-made noise rise and the path-loss exponent beyond 13 m. The assumed storey height matters least, because with

1.5 m antennas most paths pass below the roofs in any case. The field campaign was therefore designed to measure the antenna loss and the noise floor directly.

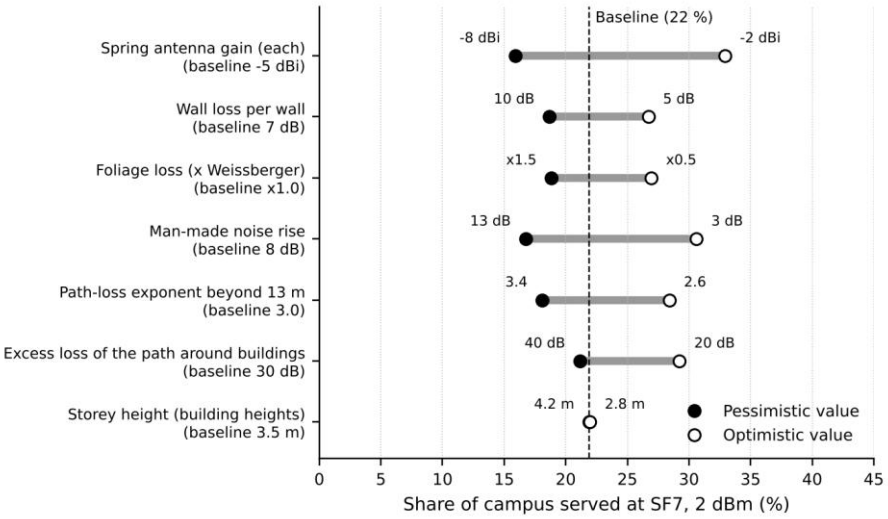


Figure 4.7: Effect of the modelling assumptions on SF7 coverage

4.1.4 Checkpoints and Low Transmit Power

Figure 4.8 divides the predicted loss to each checkpoint into distance, antenna, obstruction and foliage components. Only CP07, on the open plaza below the transmitter, has almost no obstruction loss. CP01 and CP03 have no wall in the path, but they lie about 34 m below the transmitter, so the edge of the ground in between adds more than 20 dB of diffraction loss. Most of the other checkpoints lose 20–30 dB to buildings, the range in which the paths through the walls, over the roofs and around the buildings compete. At 2 dBm the model predicts SF7 at CP07, CP09 and CP14, and no link at CP01, CP05, CP06, CP08 and CP10. This breakdown is used in Section 4.2.3 to explain the differences between the model and the measurements.

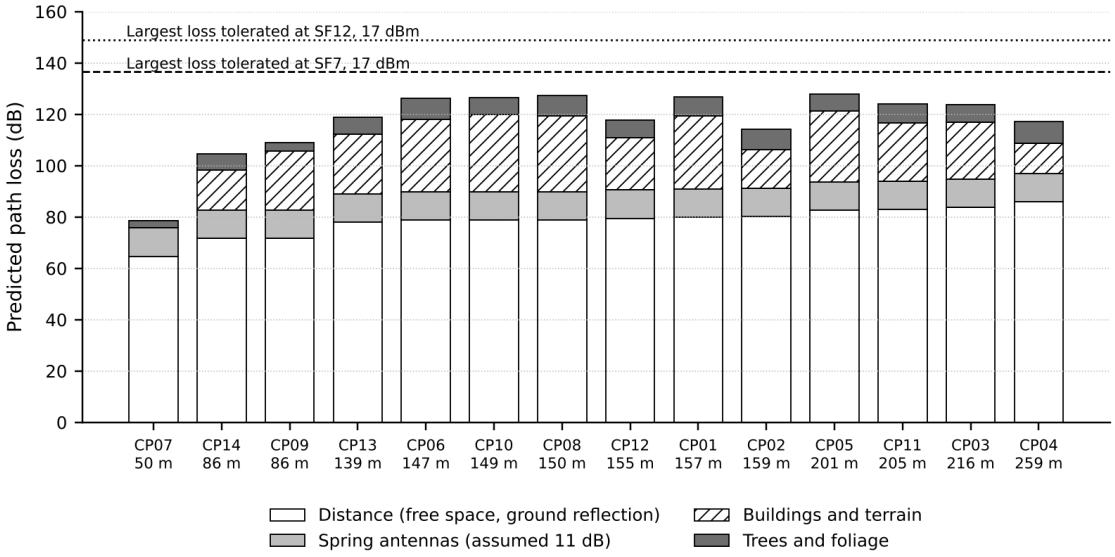


Figure 4.8: Path loss components at each checkpoint

Figure 4.9 gives the SF and transmit power with the lowest energy that still keep the PDR at or above 90 % at each checkpoint. Three checkpoints (CP07, CP09 and CP14) work at the minimum power of 2 dBm with SF7, and seven (CP01, CP03, CP05, CP06, CP08, CP10 and CP11) need the full 17 dBm. Compared with always using SF9 at 17 dBm, the median energy saving is 76 %. Increasing the power by a few decibels at SF7 costs less energy than moving to a slower SF, because each SF step doubles the time on air.

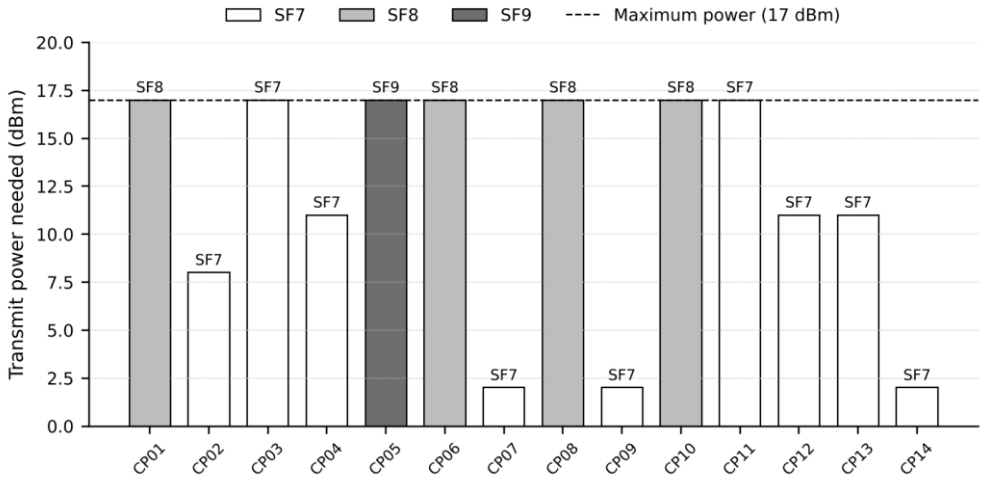


Figure 4.9: Lowest-energy SF and power at each checkpoint

4.1.5 Network Capacity

Figure 4.10 shows the network PDR as the number of nodes at random campus locations increases, with each node sending one packet every 5 minutes at 17 dBm. With 400 nodes, the per-node SF rule delivers 74 % of the packets, fixed SF9 48 % and fixed SF12 only 8

%, because the long SF12 packets collide. Selecting the SF for each node is therefore important for capacity as well as for range.

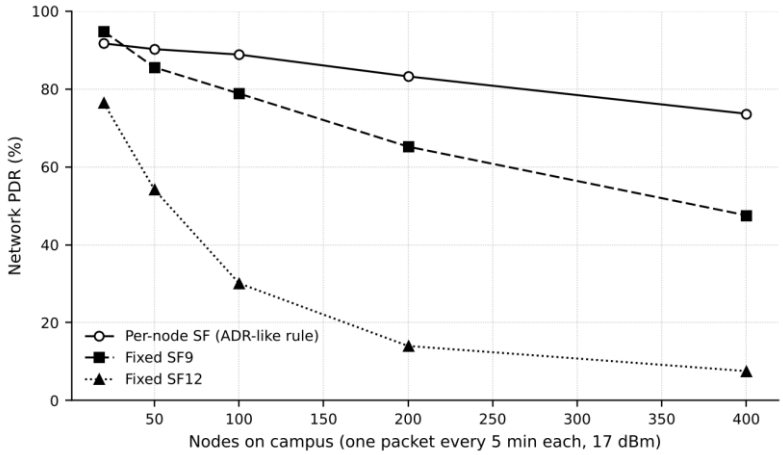


Figure 4.10: Network PDR against the number of nodes

4.1.6 Discussion

The simulation shows that the physical layer behaves as the datasheet predicts, that every parameter trades range for energy along a common curve, and that on the KU campus the lowest-energy operating point is usually SF7 at reduced power. Raising the antennas, especially the gateway, is the most effective way to extend coverage. The simulation also has limitations. The antenna gain, the wall loss, the loss of the path around buildings, the building heights and the noise level are assumptions until they are calibrated. The 30 m elevation model smooths small terrain features, and its vertical accuracy of about 4 m [28] is similar to the height of the ground rises that block some paths, so the terrain loss at individual points is uncertain. Since shadowing is averaged, the maps give the probability of meeting the target rather than the value at an exact spot, and the supply currents are datasheet values.

4.2 Field Measurement Results

The first field session was carried out on the KU campus. Five checkpoints (CP06, CP07, CP09, CP10 and CP13) were measured before rain stopped the session, giving 77 complete 20-packet windows. Figure 4.11 shows the land cover of the campus with the transmitter, the 14 simulated checkpoints and the five checkpoints measured in the field. The results are preliminary, since they come from one afternoon and one visit per checkpoint, and each result is compared with the simulation of Section 4.1. The test conditions are listed in Table 4.5.



Figure 4.11: Checkpoints on the KU campus land-cover map

Table 4.5: Field test conditions

Item	Value
Place	Kathmandu University campus, Dhulikhel
Weather and disturbances	Overcast, 19–20 °C; light rain from CP13 onwards, heavier at CP06; people walking past at CP13
Transmitter	Plaza south of the upper fountain (27.61857 N, 85.53858 E), spring antenna on a 1.5 m tripod
Receiver	Spring antenna on a 1.5 m tripod; at CP07 the tripod stood on a 0.2–0.5 m wall
Radio settings	2 dBm, BW 62.5 kHz, CR 4/7, SF7–SF12 sweep, 20 packets per setting
Checkpoints measured	CP06, CP07, CP09, CP10 and CP13; two sweeps each (CP13 and CP06 repeated after interruptions)

4.2.1 Line-of-Sight Distance Test

A line-of-sight distance test was carried out on open ground from 10 m to 180 m with SF9, BW 62.5 kHz and CR 4/7, with both antennas vertical and about 2 m above the ground. The transmit power was left at the default output of the LoRa library, which is 17 dBm. Table 4.6 lists the measured average RSSI, SNR and packet loss, together with the received power predicted by the simulation model for the same power and antenna height, and Figure 4.12 plots the same values.

Table 4.6: Line-of-sight distance test results

Distance (m)	Measured RSSI (dBm)	SNR (dB)	Packet loss (%)	Model (dBm)	Model − measured (dB)
10	-57.05	12.66	0.00	-39.2	+17.9
20	-59.80	12.64	0.00	-45.2	+14.6
30	-62.70	12.55	0.00	-49.9	+12.8
40	-66.35	12.33	0.00	-53.6	+12.7
50	-68.85	12.16	0.00	-56.5	+12.3
80	-72.65	11.61	1.00	-62.6	+10.0
95	-75.65	10.56	2.55	-64.9	+10.8
100	-77.60	9.59	3.45	-65.5	+12.1
180	-80.60	7.57	6.35	-73.2	+7.4

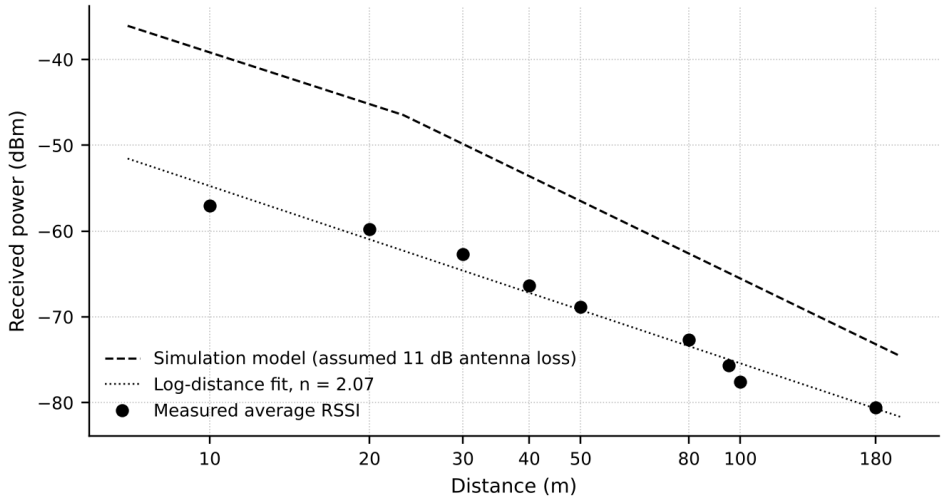


Figure 4.12: Line-of-sight RSSI against distance

The RSSI fell from -57.05 dBm at 10 m to -80.60 dBm at 180 m, and the packet loss stayed below 7 %. The measured values follow a log-distance line with a path-loss exponent of 2.07, which is close to free space. At 10 m, the model with the assumed antennas predicts -39.2 dBm. With a free-space loss of 45.2 dB at 10 m, the measured value corresponds to a total loss of about 29 dB for the two antennas and their installation, 18 dB more than the assumed 11 dB. This agrees with the 17 dB by which CP07 falls below the model in Section 4.2.3. Since the model uses an exponent of 3 beyond the breakpoint (23 m for 2 m antennas), the difference between the model and the measurement falls from 17.9 dB at 10 m to 7.4 dB at 180 m.

4.2.2 Packet Delivery at the Checkpoints

Figure 4.13 shows the measured PDR for every checkpoint and SF, with the SF predicted by the three-dimensional model boxed. Only CP07, 51 m from the transmitter on an almost clear path, delivered every packet at every SF. At CP06, CP09, CP10 and CP13 no SF reached the 90 % target; the best results were 88 % at SF12 and 85 % at SF9 at CP09, 84 % at SF12 at CP13, 78 % at SF12 at CP10, and 68 % at SF11 at CP06 in the rain. A transmit power of 2 dBm is therefore too low for most of the campus with these antennas. The delivery also varied strongly between the two sweeps at the same point, which shows the effect of people moving and of rain on a link that is close to its limit.

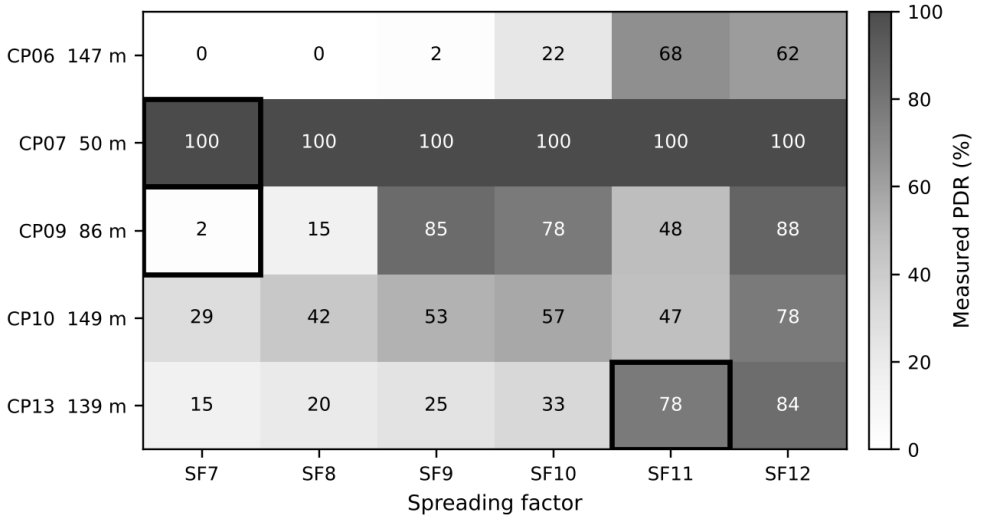


Figure 4.13: Measured PDR at each checkpoint

4.2.3 Received Power Compared with the Model

Figure 4.14 and Table 4.7 compare the median received power at each checkpoint with the three-dimensional model under the same conditions (2 dBm, 1.5 m antennas). At CP07 and CP09 the measured power is 17–18 dB lower than predicted, which points to a large loss common to all paths: most likely the spring antennas and their mounting are much worse than the assumed −5 dBi each. At CP10 and CP06, where the path crosses several buildings, the model is 6–7 dB too pessimistic. When the antenna loss is fitted to all five checkpoints, it rises from the assumed 11 dB to 18.7 dB for the pair and the root-mean-square error falls from 12.3 dB to 10.8 dB. The remaining error, with the clear paths still too strong and the blocked paths too weak in the model, suggests that the loss of the paths around the buildings is overestimated. A single fitted value therefore mixes two different errors and is only a compromise.

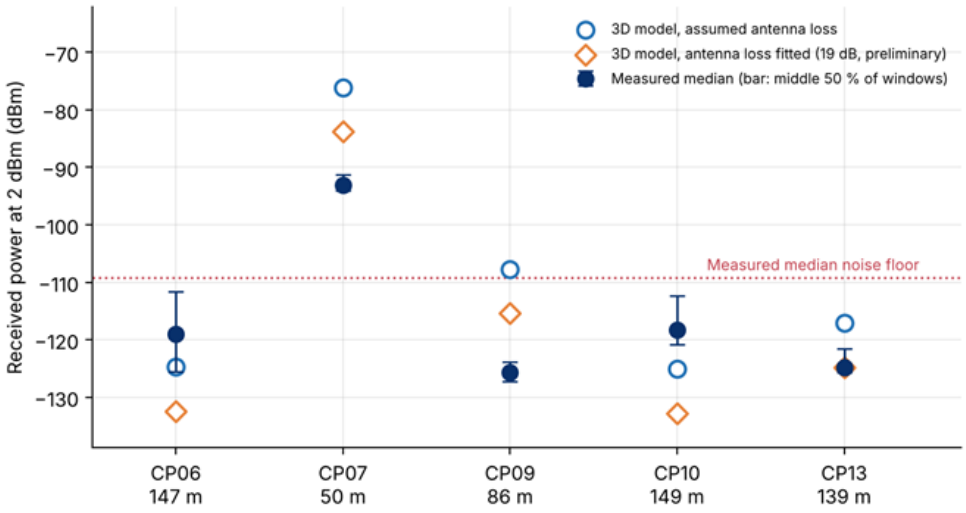


Figure 4.14: Measured and predicted received power

Table 4.7: Measured and predicted results at the checkpoints

CP	Distance (m)	Walls	Model (dBm)	Measured (dBm)	Model − measured (dB)	Predicted SF	Measured SF (≥ 90 %)
CP06	147	4	-124.7	-119.1	-5.7	none	none
CP07	51	0	-76.1	-93.1	+17.0	SF7	SF7
CP09	86	2	-107.7	-125.7	+18.0	SF7	none
CP10	149	8	-125.1	-118.3	-6.8	none	none
CP13	139	2	-117.1	-124.8	+7.7	SF11	none

Together with the line-of-sight reference of Section 4.2.1, the measurements point to three corrections to the model: the antenna and installation loss is about 18 dB larger than assumed, the path-loss slope on open ground is closer to 2 than to 3, and the loss through several buildings is overestimated. These corrections partly cancel at the obstructed checkpoints, which is why the single fitted antenna loss (18.7 dB for the pair) is smaller than the 29 dB found in line of sight. They will be calibrated together when the remaining checkpoints have been measured.

4.2.4 Receiver Sensitivity, Noise Floor and RSSI

Figure 4.15 shows the lowest window-mean SNR at which packets were still decoded. The receiver worked down to -9.6 dB at SF7 and -19.6 dB at SF12, against datasheet limits of −7.5 dB and −20 dB, which confirms the physical-layer model of Section 4.1.1. The median noise floor was -109.2 dBm, about 3 dB above the −112 dBm assumed in the model. At CP07 the RSSI stayed between −90 dBm and −95 dBm for every SF, which confirms that the received signal strength does not depend on the spreading factor.

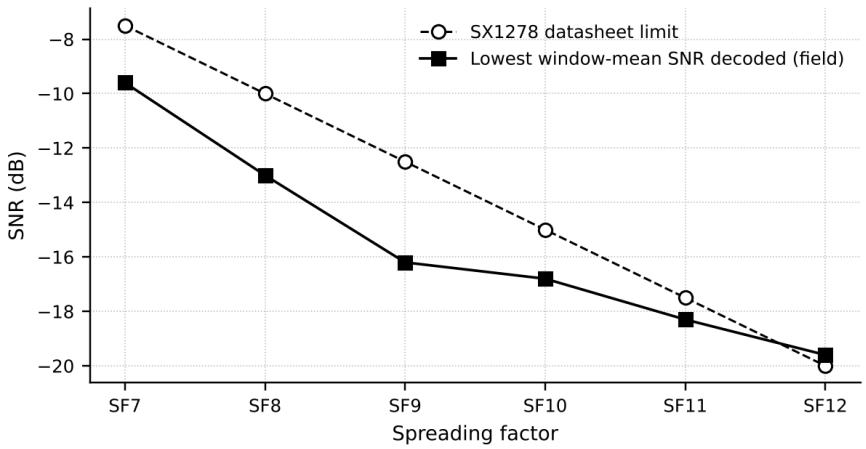


Figure 4.15: Lowest decoded SNR compared with the datasheet

4.3 Summary

The simulated chirp receiver agreed with the SX1278 datasheet limits, and all combinations of SF, BW and CR were found to lie on one range–energy curve, with the SF having the largest effect. The three-dimensional campus model predicted the lowest SF at each point from 2 dBm to 17 dBm and showed that buildings, trees and the lower western ground create the main shadow areas. Per-node SF selection improved the network capacity. In the first field session, the receiver decoded packets down to the datasheet SNR limits, but 2 dBm was too weak for most checkpoints. The antenna and installation loss was about 18 dB larger than assumed, and the path-loss slope and building loss of the model need calibration.

The 77 windows from five checkpoints are not enough to train the machine-learning models of Section 3.6 and test them on unseen checkpoints. The training programs have been checked with simulated data, and the models will be trained only on measured data. Further sessions are therefore planned at every checkpoint with transmit powers from 2 dBm to 17 dBm. The lowest SF and power giving a PDR of 90 % or more at each location will provide the training labels and the reference for adaptive SF and power control.

CHAPTER V: CONCLUSION AND FUTURE WORK

5.1 Conclusion

A simulation framework for a 433 MHz LoRa link between two ESP32 and SX1278 nodes was developed from theory and published values, together with a three-dimensional model of the Kathmandu University campus and automated firmware for field measurement. The chirp Monte-Carlo model reproduced the SX1278 demodulation limits within 0.4–1.8 dB, and the field receiver decoded packets down to the same limits. Each increase in SF doubled the time on air and energy per packet while improving the sensitivity by about 2.5 dB, and all combinations of SF, BW and CR lay on a single range–energy curve. With 1.5 m antennas, the campus model predicted that SF7 serves 60 % of the campus at 17 dBm and 22 % at 2 dBm, and selecting the lowest-energy SF and power at each checkpoint saved a median of 76 % of the energy compared with SF9 at 17 dBm. With 400 nodes, per-node SF selection maintained about 74 % delivery, against only 8 % for fixed SF12.

The first field session showed that 2 dBm is too weak for most of the campus with 1.5 m spring antennas, and that the antenna and installation loss is about 18 dB larger than assumed. In line of sight, the received power decreased close to free space (n = 2.07). The link settings and antennas therefore strongly affect both reliability and energy, and a site-specific model must be calibrated with measurements before it is used for deployment planning. These conclusions are based on one field session and remain to be confirmed with the calibrated model, the remaining field sessions and the machine-learning results.

5.2 Future Works

The future works of this project are as follows:

- The field campaign will be completed with the line-of-sight walk, the remaining checkpoints and repeat visits, using transmit powers from 2 dBm to 17 dBm.
- The three-dimensional model will be calibrated with the measured antenna loss, path-loss slope, building loss and noise floor, and better antennas will be tested.
- Machine-learning SF selectors will be trained on the measured data, compared with the LoRaWAN ADR rule and deployed with transmit-power control.
- The river water-level node will be installed at the selected site and its alerting performance verified.

GANTT CHART

Table 5.1: Gantt chart

Activity	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Literature Review	■	■	■						
Proposal Defense		■							
System Design and Hardware Planning		■	■						
LoRa Transmitter and Receiver Setup			■						
Parameter Testing (SF, BW and CR)			■	■					
Line-of-Sight Link Testing				■	■				
ESP32 Firmware for SF Sweep						■			
Field Logger and Data Analysis						■	■		
LoRa Physical-Layer and Channel Simulation						■			
3D KU Campus Model and Network Simulation						■	■		
First Field Session							■		
Remaining Field Sessions							■	■	
Model Calibration with Field Data								■	
ML Model Training and Evaluation								■	■
Adaptive SF and Power Control									■
River Node, PCB and Enclosure								■	■
Final Analysis, Report and Presentation									■

Task Completed	■
Task to be completed	■

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A. APPENDIX: HARDWARE

A.1 SX1278 (Ra-01) to ESP32 Connections

Both nodes use the same wiring between the SX1278 module and the ESP32, listed in Table A.1. The module is powered from the 3.3 V regulator of the ESP32 board.

Table A.1: SX1278 to ESP32 pin connections

Ra-01 pin	Function	ESP32 pin
3.3V / GND	Supply and common ground	3V3 / GND
SCK	SPI clock	GPIO 18
MISO	SPI data from the module	GPIO 19
MOSI	SPI data to the module	GPIO 23
NSS	SPI chip select	GPIO 5
RESET	Module reset	GPIO 14
DIO0	Transmit-done and receive-done interrupt	GPIO 26
–	Status LED	GPIO 2

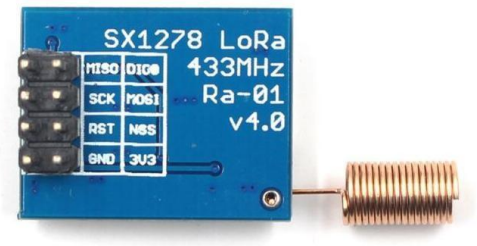


Figure A.1: SX1278 (Ra-01) 433 MHz LoRa module

A.2 Components Used

Table A.2: Components used in the project

Component	Type / rating	Qty.	Use
Microcontroller	ESP32 development board	2	Controls the LoRa module and the sensor
LoRa transceiver	Ai-Thinker Ra-01 (SX1278), 433 MHz	2	Transmitter and receiver
Antenna	Spring helical antenna supplied with the module	2	Radiating element of each node
Ultrasonic sensor	HC-SR04	1	Water-level measurement (sensor node)
Breadboard and jumper wires	–	2 sets	Circuit assembly
Tripod	Adjustable, set to 1.5 m	2	Antenna height in the field test
Laptop	USB connection to the receiver	1	Power and data logging

A.3 Hardware Setup

Figure A.1 shows the SX1278 module with its spring antenna. Figure A.2 shows the transmitting node used in the field test with its main parts labelled, and Figure A.3 shows the receiving node on its tripod at two of the checkpoints, with the laptop displaying the received packets.

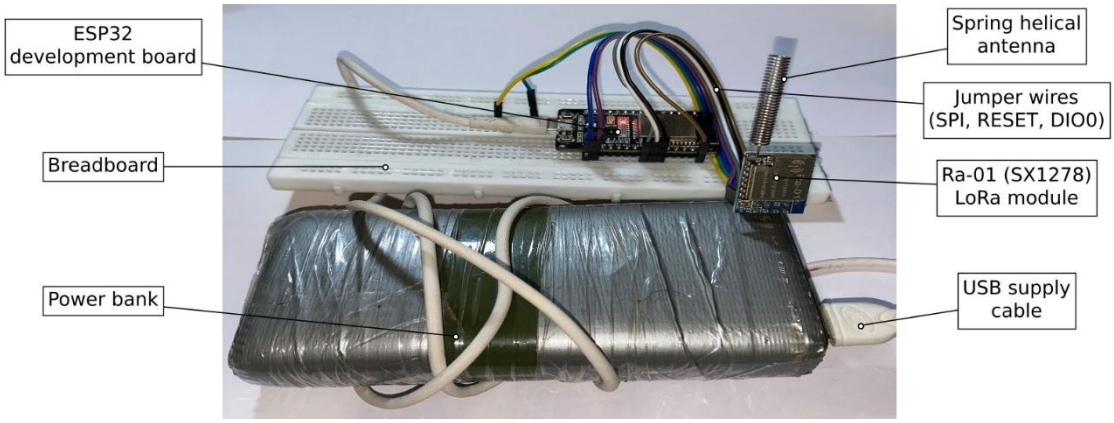
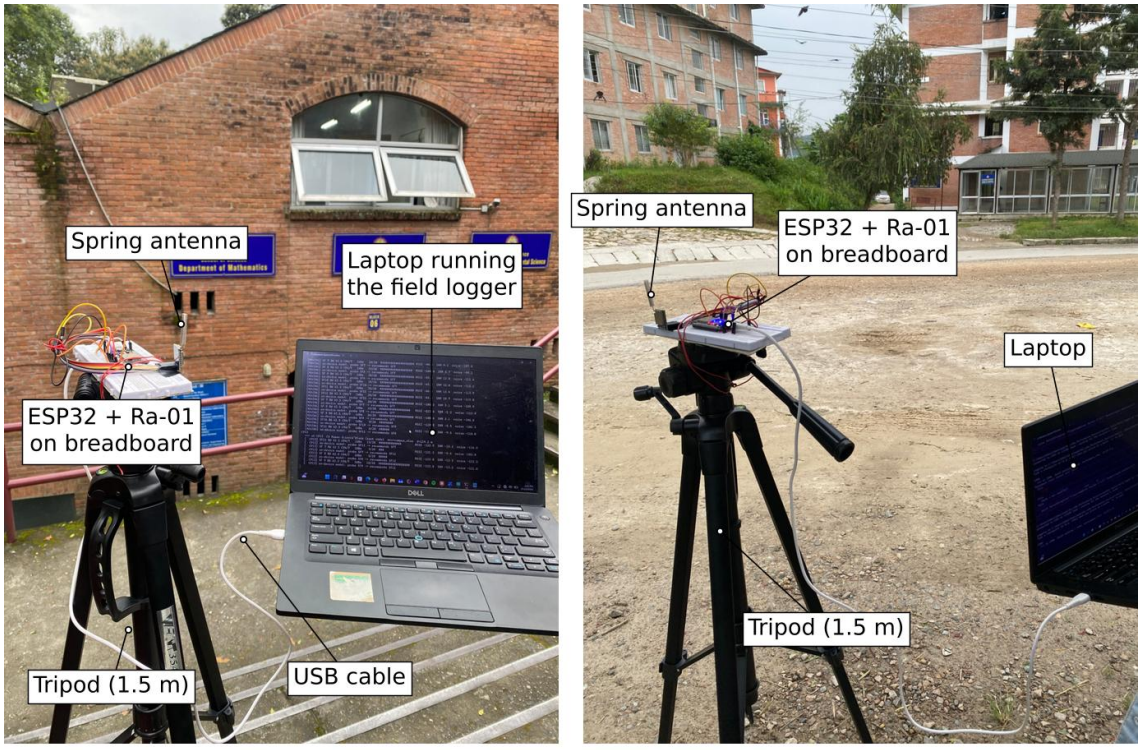


Figure A.2: Transmitting node



(a) (b)

Figure A.3: Receiving node in the field: (a) at CP13, (b) at a parking-area checkpoint

B. APPENDIX: FIELD TEST DATA

B.1 Checkpoint Locations

Table B.1 gives the location of the 14 checkpoints, the straight-line distance from the transmitter, the number of walls and the length of vegetation on the path, and the lowest SF predicted by the simulation at 17 dBm and 2 dBm.

Table B.1: Checkpoint locations and predicted lowest SF

ID	Location	Lat. (°N)	Long. (°E)	Dist. (m)	Walls	Veg. (m)	SF, 17 dBm	SF, 2 dBm
CP01	Football Ground - centre	27.61866	85.53700	157	0	70	SF8	none
CP02	Football Ground - south end	27.61817	85.53703	159	2	81	SF7	SF9
CP03	Basketball Court	27.61826	85.53642	216	0	64	SF7	SF12
CP04	West road junction (towards gate)	27.61802	85.53603	259	2	88	SF7	SF10
CP05	Octagonal hostel (west) - courtyard	27.61782	85.53673	201	4	59	SF9	none
CP06	South parking / road	27.61780	85.53736	147	4	85	SF8	none
CP07	Lower fountain plaza	27.61812	85.53867	51	0	20	SF7	SF7
CP08	South-east hostel / staff quarters	27.61771	85.53975	150	6	76	SF8	none
CP09	Central courtyard (Administration Block)	27.61935	85.53857	86	2	25	SF7	SF7
CP10	North-west blocks (Block 10 area)	27.61977	85.53795	149	8	62	SF8	none
CP11	North parking / main entrance	27.62039	85.53821	205	2	67	SF7	SF12
CP12	Multipurpose Hall - south side	27.61910	85.53712	155	2	64	SF7	SF11
CP13	CV Raman Science Block (east side)	27.61928	85.53942	139	2	56	SF7	SF11
CP14	Octagonal hostel (east)	27.61813	85.53930	86	2	52	SF7	SF7

Transmitter: paved plaza south of the upper fountain roundabout, 27.61857 °N, 85.53858 °E. "none": no SF reaches a PDR of 90 % in the simulation.

B.2 Recorded Data

Table B.2: Values recorded during the field test

Record	Values
Each packet	Sequence number, SF, BW, CR, transmit power, RSSI, SNR, frequency error, CRC status
Each 20-packet window	Packets sent and received, PDR, noise floor
Each checkpoint	Checkpoint ID, distance, environment, weather and notes

C. APPENDIX: SOFTWARE

C.1 Firmware

The firmware was written in C++ for the ESP32 with the Arduino framework and the RadioLib library [27]. Listing C.1 gives the settings and packet formats shared by both nodes, Listing C.2 the transmitter and Listing C.3 the receiver. The handling of serial commands and the optional extended sweep are not shown.

Listing C.1: Settings shared by the transmitter and the receiver (sweep_protocol.h)

```
// sweep_protocol.h - settings shared by the transmitter and the receiver

#define PIN_SCK   18
#define PIN_MISO  19
#define PIN_MOSI  23
#define PIN_NSS    5
#define PIN_RST   14
#define PIN_DIO0  26
#define PIN_LED    2

#define RF_FREQ_MHZ   433.175
#define SYNC_WORD     0x12
#define PREAMBLE_LEN  8
#define DATA_PAYLOAD  20          // bytes per measurement packet

// control channel used only for the announce packets
#define CTRL_SF   12
#define CTRL_BW   125.0
#define CTRL_CR   8
#define CTRL_PWR  17

#define LEAD_MS           400     // announce -> first data packet
#define GAP_MS            200     // added to the time on air between packets
#define WINDOW_MARGIN_MS  800

#define MAGIC_ANN   0x41      // 'A'
#define MAGIC_DATA  0x44      // 'D'

typedef struct __attribute__((packed)) {
  uint8_t  magic;                 // 'A'
  uint8_t  beacon_id;
  uint16_t sweep_id;
  uint8_t  cfg_id;
  uint8_t  sf, bw_code, cr;       // bw_code: 0 = 62.5, 1 = 125, 2 = 250 kHz
  int8_t   pwr_dbm;
  uint8_t  n_pkts;
  uint16_t interval_ms, lead_ms;
  uint8_t  payload_len;
} AnnouncePkt;

typedef struct __attribute__((packed)) {
  uint8_t  magic;                 // 'D'
  uint8_t  beacon_id;
  uint16_t sweep_id;
  uint8_t  cfg_id, seq, n_pkts;
  int8_t   pwr_dbm;
  uint32_t prev_energy_uJ;
  uint8_t  filler[DATA_PAYLOAD - 12];
} DataPkt;

static const float BW_TABLE[4] = {62.5, 125.0, 250.0, 500.0};
inline uint8_t bwToCode(float bw) {
  for (uint8_t i = 0; i < 4; i++) if (fabs(BW_TABLE[i] - bw) < 1) return i;
  return 1;
}
```

```
typedef struct { uint8_t sf; float bw; uint8_t cr; int8_t pwr; } RadioCfg;
```

Listing C.2: Transmitter: automatic SF sweep (beacon_tx.ino)

```
// beacon_tx.ino - transmitter: steps through SF7 to SF12 and sends
// 20 numbered packets at each setting
#include <RadioLib.h>
#include <SPI.h>
#include "sweep_protocol.h"

#define BEACON_ID 1

SX1278 radio = new Module(PIN_NSS, PIN_DIO0, PIN_RST, RADIOLIB_NC);

int8_t   campaignPwr = 2;         // dBm, changed with the serial command p<dBm>
uint8_t  nPkts = 20;
uint16_t sweepId = 0;
RadioCfg cfgs[24];
uint8_t  nCfg = 0;

volatile bool txDone = false;
void IRAM_ATTR onTxDone() { txDone = true; }

void buildSweep() {
  nCfg = 0;
  for (uint8_t sf = 7; sf <= 12; sf++)
    cfgs[nCfg++] = {sf, 62.5, 7, campaignPwr};   // BW 62.5 kHz, CR 4/7
}

bool applyCfg(uint8_t sf, float bw, uint8_t cr, int8_t pwr) {
  int s = radio.setSpreadingFactor(sf);
  s |= radio.setBandwidth(bw);
  s |= radio.setCodingRate(cr);
  s |= radio.setOutputPower(pwr);
  return s == RADIOLIB_ERR_NONE;
}

uint32_t toaMs(size_t len) { return (radio.getTimeOnAir(len) + 999) / 1000; }

void sendPacket(uint8_t *buf, size_t len) {
  txDone = false;
  radio.startTransmit(buf, len);
  uint32_t t0 = millis();
  while (!txDone && millis() - t0 < 6000) delay(1);
  radio.finishTransmit();
  digitalWrite(PIN_LED, !digitalRead(PIN_LED));
}

void sendAnnounce(uint8_t cfgId, const RadioCfg &c, uint16_t interval) {
  applyCfg(CTRL_SF, CTRL_BW, CTRL_CR, CTRL_PWR);
  AnnouncePkt a = {MAGIC_ANN, BEACON_ID, sweepId, cfgId, c.sf, bwToCode(c.bw),
                   c.cr, c.pwr, nPkts, interval, 0, DATA_PAYLOAD};
  a.lead_ms = LEAD_MS + 150 + toaMs(sizeof(a));  // first copy also covers the second
  sendPacket((uint8_t *)&a, sizeof(a));
  delay(150);
  a.lead_ms = LEAD_MS;
  sendPacket((uint8_t *)&a, sizeof(a));
}

void runConfig(uint8_t cfgId, const RadioCfg &c) {
  applyCfg(c.sf, c.bw, c.cr, c.pwr);
  uint16_t interval = toaMs(DATA_PAYLOAD) + GAP_MS;
  sendAnnounce(cfgId, c, interval);
  delay(LEAD_MS);
  applyCfg(c.sf, c.bw, c.cr, c.pwr);

  DataPkt d = {};
  d.magic = MAGIC_DATA; d.beacon_id = BEACON_ID;
  d.sweep_id = sweepId; d.cfg_id = cfgId;
  d.n_pkts = nPkts; d.pwr_dbm = c.pwr;
  for (uint8_t s = 0; s < nPkts; s++) {
    uint32_t t = millis();
    d.seq = s;
    sendPacket((uint8_t *)&d, sizeof(d));
```

```
    while (millis() - t < interval) { handleSerial(); delay(1); }
  }
}

// handleSerial() reads the commands p<dBm> and n<N>; it is not shown here.

void setup() {
  Serial.begin(115200);
  pinMode(PIN_LED, OUTPUT);
  SPI.begin(PIN_SCK, PIN_MISO, PIN_MOSI, PIN_NSS);
  int st = radio.begin(RF_FREQ_MHZ, CTRL_BW, CTRL_SF, CTRL_CR,
                       SYNC_WORD, CTRL_PWR, PREAMBLE_LEN);
  if (st != RADIOLIB_ERR_NONE) {
    Serial.printf("ERR,begin,%d\n", st);
    while (true) { digitalWrite(PIN_LED, !digitalRead(PIN_LED)); delay(100); }
  }
  radio.setCRC(true);
  radio.setCurrentLimit(140);     // needed for +17 dBm
  radio.setPacketSentAction(onTxDone);
  buildSweep();
}

void loop() {
  handleSerial();
  sweepId++;
  Serial.printf("SWEEP,%u,start,ms=%lu\n", sweepId, millis());
  for (uint8_t i = 0; i < nCfg; i++) runConfig(i, cfgs[i]);
}
```

Listing C.3: Receiver: following the sweep and logging each packet (mobile_rx.ino)

```
// mobile_rx.ino - receiver: follows the transmitter's settings and sends
// every packet and every 20-packet window to the laptop over USB
#include <RadioLib.h>
#include <SPI.h>
#include "sweep_protocol.h"

SX1278 radio = new Module(PIN_NSS, PIN_DIO0, PIN_RST, RADIOLIB_NC);

volatile bool rxFlag = false;
void IRAM_ATTR onRx() { rxFlag = true; }

enum State { ST_CONTROL, ST_DATA };
State state = ST_CONTROL;
AnnouncePkt cur;
uint32_t windowEnd = 0;
uint8_t  nRx = 0, nCrc = 0;
float    noiseDbm = 0;
uint16_t lastSweep = 0xFFFF;
uint8_t  lastCfg = 0xFF;

bool setRadio(uint8_t sf, float bw, uint8_t cr) {
  return (radio.setSpreadingFactor(sf) | radio.setBandwidth(bw) |
          radio.setCodingRate(cr)) == RADIOLIB_ERR_NONE;
}

float measureNoise() {            // RSSI with no packet on air
  radio.startReceive();
  delay(3);
  float acc = 0;
  for (int i = 0; i < 8; i++) { acc += radio.getRSSI(false, true); delay(2); }
  return acc / 8.0;
}

void listenControl() {
  setRadio(CTRL_SF, CTRL_BW, CTRL_CR);
  state = ST_CONTROL;
  radio.startReceive();
}

void startWindow(const AnnouncePkt &a) {
  cur = a;
  float bw = BW_TABLE[a.bw_code & 3];
  setRadio(a.sf, bw, a.cr);
  noiseDbm = measureNoise();
```

```
  nRx = nCrc = 0;
  windowEnd = millis() + a.lead_ms + (uint32_t)a.n_pkts * a.interval_ms
              + WINDOW_MARGIN_MS;
  state = ST_DATA;
  radio.startReceive();
  Serial.printf("ANN,%u,%u,%u,%.1f,%u,%d,%u,%u\n", a.sweep_id, a.cfg_id,
                a.sf, bw, a.cr, a.pwr_dbm, a.n_pkts, a.interval_ms);
}

void endWindow() {
  Serial.printf("WIN,%u,%u,%u,%.1f,%u,%d,%u,%u,%u,%.1f\n", cur.sweep_id,
                cur.cfg_id, cur.sf, BW_TABLE[cur.bw_code & 3], cur.cr,
                cur.pwr_dbm, cur.n_pkts, nRx, nCrc, noiseDbm);
  listenControl();
}

void onPacket() {
  uint8_t buf[64];
  size_t len = min((size_t)radio.getPacketLength(), sizeof(buf));
  int st = radio.readData(buf, len);
  float rssi = radio.getRSSI(), snr = radio.getSNR();
  float ferr = radio.getFrequencyError();

  if (state == ST_CONTROL) {
    if (st == RADIOLIB_ERR_NONE && len == sizeof(AnnouncePkt) && buf[0] == MAGIC_ANN) {
      AnnouncePkt a;
      memcpy(&a, buf, sizeof(a));
      if (a.sweep_id != lastSweep || a.cfg_id != lastCfg) {   // skip the repeat
        lastSweep = a.sweep_id; lastCfg = a.cfg_id;
        startWindow(a);
        return;
      }
    }
    radio.startReceive();
    return;
  }

  if (st == RADIOLIB_ERR_CRC_MISMATCH) {
    nCrc++;
  } else if (st == RADIOLIB_ERR_NONE && len == sizeof(DataPkt) && buf[0] == MAGIC_DATA)
{
    DataPkt d;
    memcpy(&d, buf, sizeof(d));
    if (d.sweep_id == cur.sweep_id && d.cfg_id == cur.cfg_id) {
      nRx++;
      Serial.printf("PKT,%u,%u,%u,%u,%.1f,%u,%d,%.1f,%.2f,%.0f,1\n",
                    d.sweep_id, d.cfg_id, d.seq, cur.sf, BW_TABLE[cur.bw_code & 3],
                    cur.cr, d.pwr_dbm, rssi, snr, ferr);
    }
  }
  radio.startReceive();
}

void setup() {
  Serial.begin(115200);
  SPI.begin(PIN_SCK, PIN_MISO, PIN_MOSI, PIN_NSS);
  radio.begin(RF_FREQ_MHZ, CTRL_BW, CTRL_SF, CTRL_CR, SYNC_WORD, 10, PREAMBLE_LEN);
  radio.setCRC(true);
  radio.setPacketReceivedAction(onRx);
  listenControl();
}

void loop() {
  if (rxFlag) { rxFlag = false; onPacket(); }
  if (state == ST_DATA && (int32_t)(millis() - windowEnd) > 0) endWindow();
}
```

C.2 Simulation Software

The simulation was written in Python with NumPy, SciPy and pandas. Listings C.4 to C.7 give the main functions. Constants written in capitals, such as LAM, FSPL_1M and L_WALL_DB, are read from the configuration file and have the values of Table 3.1.

Listing C.4: Chirp modulation and demodulation (css_montecarlo.py)

```
# css_montecarlo.py - chirp modulation, demodulation and symbol error rate
import numpy as np


def base_upchirp(sf):
    m = 2 ** sf
    n = np.arange(m)
    return np.exp(1j * np.pi * (n * n / m - n))


def modulate(symbols, sf):
    m = 2 ** sf
    n = np.arange(m)
    k = (n[None, :] + symbols[:, None]) % m          # cyclic shift by the symbol
    return np.exp(1j * np.pi * (k * k / m - k))      # Equation (1)


def demodulate(rx, sf):
    down = np.conj(base_upchirp(sf))
    spec = np.abs(np.fft.fft(rx * down[None, :], axis=1))
    return np.argmax(spec, axis=1)                   # Equation (2)


def ser_point(sf, snr_db, n_sym, rng):
    m = 2 ** sf
    sym = rng.integers(0, m, n_sym)
    x = modulate(sym, sf)
    sigma = np.sqrt(10 ** (-snr_db / 10) / 2)
    noise = sigma * (rng.normal(size=x.shape) + 1j * rng.normal(size=x.shape))
    return np.mean(demodulate(x + noise, sf) != sym)
```

Listing C.5: Time on air, energy and packet error (lora_phy.py)

```
# lora_phy.py - time on air, energy and packet error
import math
import numpy as np


def symbol_time_s(sf, bw_khz):
    return 2 ** sf / (bw_khz * 1e3)


def n_payload_symbols(c):
    de = 1 if symbol_time_s(c.sf, c.bw_khz) > 16e-3 else 0
    ih = 0 if c.explicit_header else 1
    crc = 1 if c.crc else 0
    num = 8 * c.payload_bytes - 4 * c.sf + 28 + 16 * crc - 20 * ih
    den = 4 * (c.sf - 2 * de)
    return 8 + max(math.ceil(num / den) * (c.cr + 4), 0)        # Equation (4)


def time_on_air_s(c):
    ts = symbol_time_s(c.sf, c.bw_khz)
    return (c.preamble + 4.25) * ts + n_payload_symbols(c) * ts  # Equation (5)


def energy_per_packet_mj(c):
    i_ma = tx_current_ma(c.tx_power_dbm) + ESP32_ACTIVE_MA
    return SUPPLY_V * i_ma * time_on_air_s(c)                    # Equation (12)


def per_from_snr(c, snr_db):
    ps = ser_awgn(c.sf, np.asarray(snr_db, float) - IMPLEMENTATION_LOSS_DB)
    blk = 4 + c.cr
    n_blk = max(1, round(n_payload_symbols(c) / blk))
    if c.cr >= 3:     # CR 4/7 and 4/8 correct one symbol error per block
        p_ok = (1 - ps) ** blk + blk * ps * (1 - ps) ** (blk - 1)
    else:
        p_ok = (1 - ps) ** blk
    return 1.0 - p_ok ** n_blk                                   # Equation (10)
```

Listing C.6: Path loss on the three-dimensional campus model (ku3d.py)

```
# ku3d.py - path loss on the 3D campus model
import numpy as np


def dual_slope_db(d_m, ht, hr):
    d = np.maximum(d_m, 1.0)
    bp = np.maximum(4 * ht * hr / LAM, 1.0)                      # breakpoint
    return (FSPL_1M + 10 * N1 * np.log10(np.minimum(d, bp))
            + 10 * N2 * np.log10(np.maximum(d / bp, 1.0)))       # Equation (7)
```

```
def knife_edge_db(v):
    v = np.asarray(v, float)
    return np.where(v > -0.78,
                    6.9 + 20 * np.log10(np.sqrt((v - 0.1) ** 2 + 1) + v - 0.1),
                    0.0)                                          # Equation (16)

def weissberger_db(d_m, f_ghz=0.433):
    d = np.asarray(d_m, float)
    out = np.where(d <= 14, 0.45 * f_ghz ** 0.284 * d,
                   1.33 * f_ghz ** 0.284 * np.maximum(d, 1e-9) ** 0.588)
    return np.where(d > 0, out, 0.0)

def obstruction_db(walls, inside_m, L_dif, blocked):
    L_pen = walls * L_WALL_DB + ALPHA_IN_DB_M * inside_m
    p = 10 ** (-L_dif / 10) + 10 ** (-SCATTER_DB / 10)
    p = np.where(blocked, p + 10 ** (-L_pen / 10), p)
    return np.maximum(-10 * np.log10(p), 0.0)                     # Equation (15)

def total_loss_db(d3_m, ht, hr, walls, inside_m, L_dif, blocked, veg_m):
    return (dual_slope_db(d3_m, ht, hr) + ANTENNA_LOSS_DB
            + obstruction_db(walls, inside_m, L_dif, blocked)
            + weissberger_db(VEG_TREE_FRACTION * veg_m))          # Equation (14)
```

Listing C.7: Antenna loss fitted to the field measurements (compare_field_3d.py)

```
# compare_field_3d.py - antenna loss fitted to the measured checkpoints
import numpy as np
import pandas as pd
from scipy.optimize import least_squares

d = pd.read_csv("results/field_day1_vs_3d.csv")
meas = d.meas_median_dbm.values
base = d.pred3d_dbm.values + 11.0       # model power without the antenna loss

# one common antenna loss for all checkpoints; soft_l1 limits the pull
# of a single checkpoint that disagrees strongly
fit = least_squares(lambda p: base - p[0] - meas, [11.0], loss="soft_l1")
loss = fit.x[0]
rms_before = np.sqrt(np.mean((base - 11.0 - meas) ** 2))
rms_after = np.sqrt(np.mean((base - loss - meas) ** 2))
print(f"antenna loss {loss:.1f} dB, RMS {rms_before:.1f} -> {rms_after:.1f} dB")
```